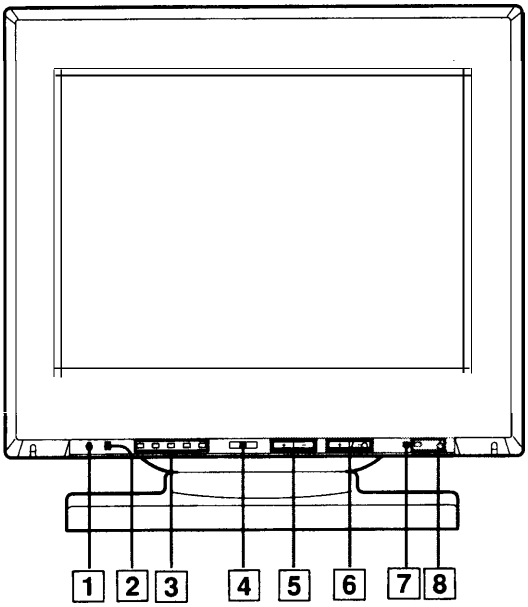
SONY MODEL 20E01

Function of Controls

See the given pages for further description.



1 RESET **button (page 9)**
Press to reset the adjustment data to the factory-preset levels for the input signal being received.

2 **CTRL (control) button (page 9)**
To reset all the adjustment data to the factory-preset levels, press the RESET button 4 while holding down this button.

3 **Indicator (page 7, 8, 9)**
The indicator corresponding to the item selected with the SELECT button 4 lights up.
CENT: Centering
SIZE: Picture size
GEOM: Picture tilt/Horizontal pincushion
CONV: Convergence
C TEMP: Color temperature

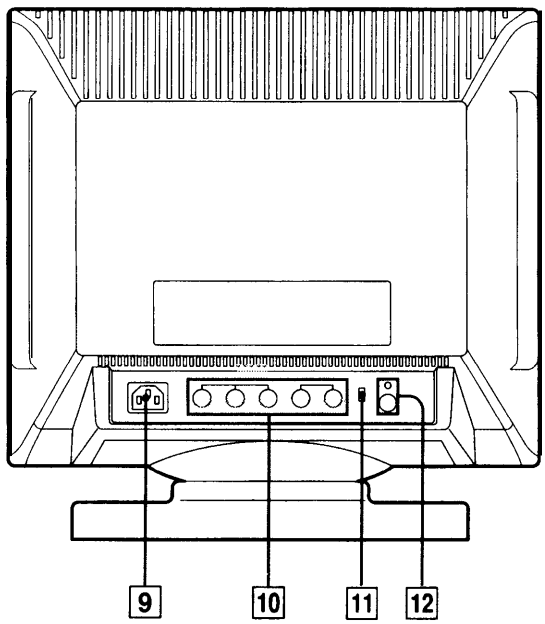
4 SELECT **button (page 7, 8, 9)**
Press to select an item.
Hold down for more than 1 second until an indicator lights up, then press repeatedly to select an item 3.

5 ☼ **(brightness) buttons (page 7, 8, 9)**
Press + or – to adjust the picture brightness.
When an item is selected, use these buttons to adjust the picture size and centering in the vertical direction and the picture tilt.

6 **◑ (contrast) buttons (page 7, 8)**
Press + or -to adjust the picture contrast.
When an item is selected, use these buttons to adjust the picture size and centering in the horizontal direction, the horizontal pincushion and the colour temperature.

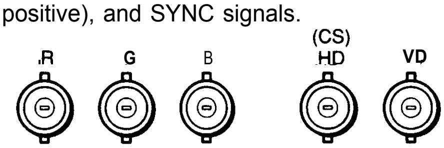
7 POWER SAVING **indicator**
Lights up when the monitor is in Power Saving Mode. Blinks when you reached the limit of the adjustable range or while in resetting mode.

8 **⏻ power switch and indicator**
Press to turn the monitor on or off. The indicator lights up when the monitor is turned on. To disconnect the power supply, disconnect the AC power cord.



[9] AC IN connector
Plug in an AC power cord.

[10] Video input connector (5 BNC)
The connector accepts RGB video signals **(0.714 Vp-p,** positive), and SYNC signals.



Note

If you use a computer or video board of high output level (about 1.0 Vp-p), you may not be able to obtain the optimum display. In such case, try decreasing the picture contrast, or use a computer or video board with a lower output level.

[11] 75Ω/2 kΩ selector
Switches the impedance of the video input connector ❑ termination to 75Ω or to 2 kΩ.

[12] SERVICE button and terminal
The button and the terminal are to be used by service personnel only.

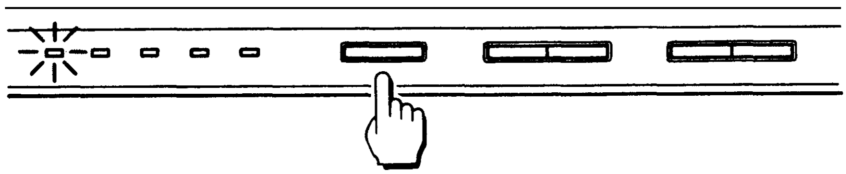
Adjustment

When one of the preset-type signals is input, no picture adjustment is necessary.
You can, however, adjust the picture to your preferences by following the procedure described below.
You can store up to 5 settings.

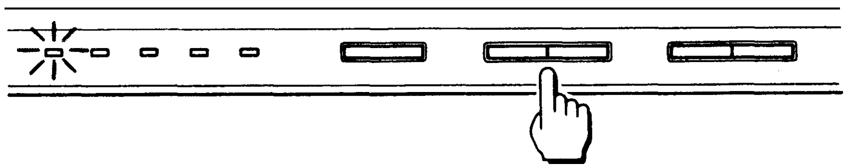
Procedure

1 **Turn on the unit, and feed the video signal from the connected computer/work station.**

2 **Press the SELECT button to choose the adjustment item.**
Pressing the SELECT button for about 1 second turns on the CENT indicator. Each time you press the SELECT button, the item changes in the following order:
CENT → SIZE → GEOM → CONV → C TEMP
For what the indicators mean and how to make an adjustment, see the following pages.



3 **Observe the picture, and press ☼ +/-, ◑ +/– buttons while the target indicator is lit.**

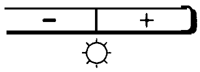


If you do not press the SELECT button or ☼ +/-, ◑ +/– buttons for more than 8 seconds, the indicator turns off.

If you need to adjust the monitor for the input signal, follow the procedures below. The adjustment data will be stored in the memory 2 seconds after you finish making the adjustments.

Adjusting the Picture Brightness

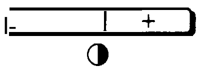
The adjustment data becomes the common setting for all input signals.



Press ☼ +/– buttons.
+ . . . for more brightness
– . . . for less brightness

Adjusting the Picture Contrast

The adjustment data becomes the common setting for all input signals.

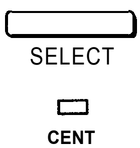


Press ◑ +/– buttons.
+ . . . for more contrast
– . . . for less contrast

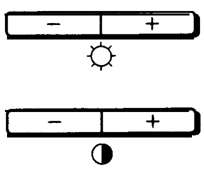
Adjusting the Centering

The adjustment data becomes the unique setting for the input signal received.

1



2



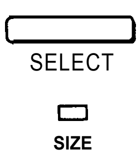
1 Press the SELECT button to turn on the CENT indicator

2 Adjust the vertical direction using ☼ +/-.
+ ... to move up
- ... to move down
Adjust the horizontal direction using ◑ +/-.
+ ... to move to the right
- ... to move to the left

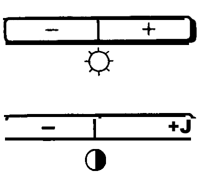
Adjusting the Picture Size

The adjustment data becomes the unique setting for the input signal received.

1



2



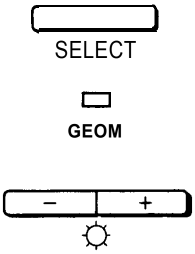
1 Press the SELECT button to turn on the SIZE indicator.

2 Adjust the vertical direction using ☼ +/-.
Adjust the horizontal direction using ◑ +/-.
+ . . . to enlarge
- . . . to diminish

Adjusting the Picture Tilt

The adjustment data becomes the common setting for all input signals.

1 2



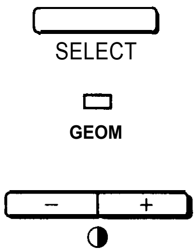
1 Press the SELECT button to turn on the GEOM indicator.

2 Correct the picture tilt using the ☼ +/– button.
+ . . . to tilt clockwise
– . . . to tilt counterclockwise

Adjusting the Horizontal Pincushion

The adjustment data becomes the unique setting for the input signal received.

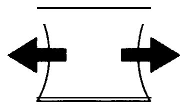
1 2



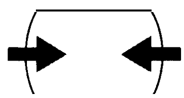
1 Press the SELECT button to turn on the GEOM indicator.

2 Correct the horizontal pincushion using ◑ +/– button.

+ . . . to expand the picture sides



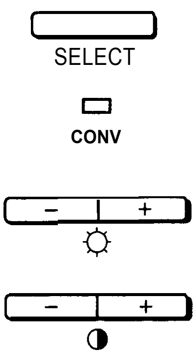
– . . . to diminish the picture sides



Adjusting the Convergence

The adjustment data becomes the common setting for all input signals.

1 2

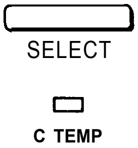


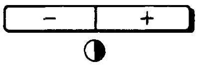
1 Press the SELECT button to turn on the CONV indicator.

2 Adjust the vertical direction using ☼ +/-.
+ . . . to move Red up and Blue down
. . to move Red down and Blue up
Adjust the horizontal direction using ◑ +/-.
+ . . . to move Red to the right and Blue to the left
– . . . to move Red to the left and Blue to the right

Selecting the Color Temperature

The selected color temperature becomes the common setting for all input signals.

1 

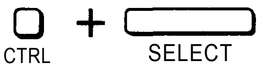
2 

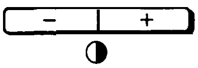
1 Press the SELECT button to turn on the C TEMP indicator.
The indicator shows the present color temperature as follows.
Lights on: 9300 K + 8 MPCD
Slow blink: 6500 K + 8 MPCD (D65)
Fast blink: 5000 K + 8 MPCD

2 Select the desired color temperature using ◑ +/– buttons.

When the background of the picture shows up

If the background of the picture is green, or the brightness does not change even when you try to adjust it using the ☼ +/- buttons, set the clamp pulse to obtain the correct black level.

1 

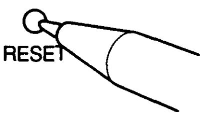
2 

1 While holding down the CTRL button, press the SELECT button for 2 seconds.
The indicators except CENT light up.

2 Set the clamp pulse so that the picture becomes normal using the ◑ +/– buttons.

Resetting the Adjustment Data to Factory-preset Levels

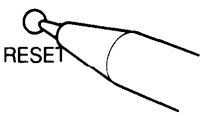
To reset the brightness and contrast adjustment data



Press the RESET button.

To reset the adjustment data of individual input signal

2 

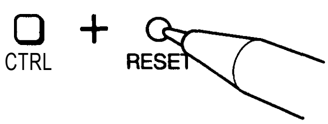
3 

1 Input the signal whose data you want to return to factory-preset level.

2 Select any one of five adjustment items (picture size, centering, convergence, tilt, horizontal pincushion) with the SELECT button.

3 Press the RESET button.

To reset all adjustment data to factory-preset levels



While holding down the CTRL button, press the RESET button.
All adjustment data including the brightness and contrast are reset to factory-preset levels.

Power Saving Function

The unit supports the computers corresponding to the " Power Saving Function" .

Power saving operation

- The H sync or V sync is not present.
  ↓
  The unit goes into suspend mode.

- Both the H sync and V sync are not present.
  ↓
  The unit goes into active-off mode.

Note

As this unit has the Power Saving Function, if you turn on the unit with no signal input, the unit goes into Power Saving mode and waits for an input signal.

LED Indicators

This table illustrates the status of the LED indicators.

	Power Saving Function			Failure Indicator
	Normal Operation	Suspend	Active-off	
⏻ Power Indicator	green	green	Off	flash
Power Saving Indicator	Off	orange	orange	flash

- If both indicators are flashing, please trip the power switch once. If they are still flashing, there is a potential monitor failure. Please note which function indicator is lit, unplug the AC cord, and contact your dealer.

Specifications

GDM-17E01

Picture tube: 0.26 mm phosphor trio pitch
17 inches measured diagonally
(16" visual)
90-degree deflection

Effective picture size
Approx. 328 x 242 mm (w/h)
(13 x 9-5/8 inches)

Resolution: Horizontal: Max. 1280 dots
Vertical: Max. 1024 lines

Display picture size
Approx. 293 x 234 mm (w/h)
(11-5/8 x 9-1/4 inches

Approx. 300 x 225 mm (w/h)
(11-7/8 x 8-7/8 inches)

Deflection frequency
Horizontal: 29 to 82 kHz
Vertical: 50 to 150 Hz

AC input voltage/current
100 to 120 V, 50/60 Hz, 2.7A
220 to 240V, 50-60Hz, 1.5A

Dimensions: 406 x 414 x 450mm
(16 x 16-3/8 x 17-3/4 inches)

Mass: Approx. 22.5 kg (49 lb 8 oz)

GDM-20E01

Picture tube: 0.31 mm phosphor trio pitch
20 inches measured diagonally
(19" visual)
90-degree deflection

Effective picture size
Approx. 388 x 292 mm (w/h)
(15 x 11-1/2 inches)

Resolution: Horizontal: Max. 1280 dots
Vertical: Max. 1024 lines

Display picture size
Approx. 350 x 280 mm (w/h)
(13-7/8 x 11-1/8 inches)

Approx. 360 x 270 mm (w/h)
(14-1/4 x 10-3/4 inches)

Deflection frequency
Horizontal: 29 to 82 kHz
Vertical: 50 to 150 Hz

AC input voltage/current
100 to 120 V, 50/60 Hz, 3.0 A
220 to 240 V, 50 – 60 Hz, 1.8 A

Dimensions: 474 x 474 x 501.5 mm
(18-3/4 x 18-3/4 x 19-3/4 inches)

Mass: Approx. 30 kg (66 lb 2 oz)

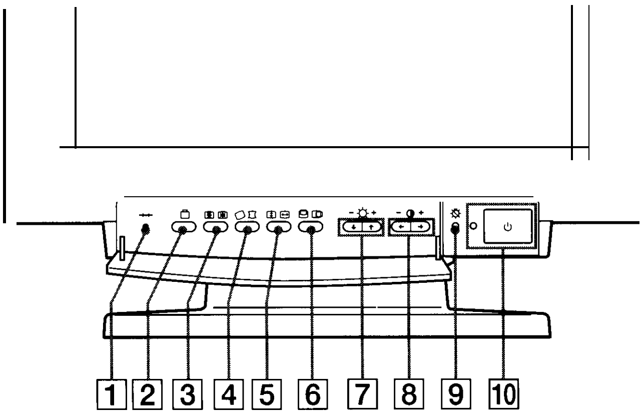
Design and specifications are subject to change without notice.

SONY MODEL 20E03

Functions of Controls

See the given pages for further description.

Front



1 ↠↞ **(reset) button (page 9)**
Press to reset the adjustment data to the factory-preset levels.

2 ▭ **(option) button (page 7-9)**
Press to set the option OSD (On Screen Display).

3 **(convergence) button (page 7)**
Press to adjust the vertical and horizontal convergence.

4 **(geometory) button (page 6)**
Press to adjust the rotation and pincushion.

5 **(size) button (page 6)**
Press to adjust the vertical and horizontal picture size.

6 **(center) button (page 5)**
Press to adjust the vertical and horizontal picture position.

7 ☼ **(brightness) −/+ (↓/↑) buttons (page 5-8)**
Press to adjust the brightness.

Act as the −/+(↓/↑) buttons to adjust the other items.

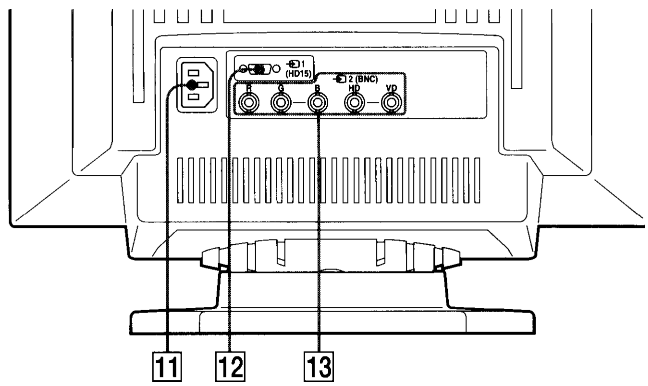
8 ◑ **(contrast) −/+ (←/→) buttons (page 5-8)**
Press to adjust the contrast.

Act as the −/+(←/→) buttons to adjust the other items.

9 **power saving indicator (page 10)**
Lights up when the monitor is in the Power Saving Modes.

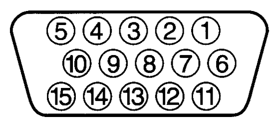
10 ⏻ **power switch and indicator (page 10)**
Press to turn the monitor on or off. The indicator lights up when the monitor is turned on.

Rear



11 **AC IN connector**
Plug in an AC power cord.

12 **Video input 1 connector (HD15)**
The connector accepts RGB video signals (0.714 Vp-p, positive) and SYNC signals.



Pin No.	Signal	Pin No.	Signal
1	Red	8	Blue Ground
2	Green (Composite Sync on Green)	9	DDC + 5V*
		10	Ground
		11	—
3	Blue	12	Bi-Directional Data (SDA)*
4	—		
5	DDC Ground*	13	H. Sync
6	Red Ground	14	v.svnc
7	Green Ground	15	Data Clock(SCL)*

* Display Data Channel (DDC) Standard by VESA

13 **Video input 2 connector (5 BNC)**
The connector accepts RGB video signals (0.714 Vp-p, positive).

Adjustments

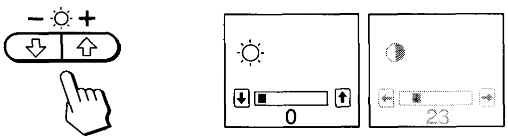
You can adjust the picture to your preferences by following the procedure described below.
You can adjust all the items on the OSD. The item to be adjusted is indicated white on the OSD.

Before adjusting the items, turn on the unit and feed the video signal from the connected computer/work station.

Adjusting the Picture Brightness

The adjustment data becomes the common setting for all input signals received.

1 Press the ☼⬇/⬆ button.
The ☼/◑ OSD (On Screen Display) appears.



2 Press the ☼⬇/⬆ buttons to adjust picture brightness.
⬇ ... for less brightness
⬆ ... for more brightness

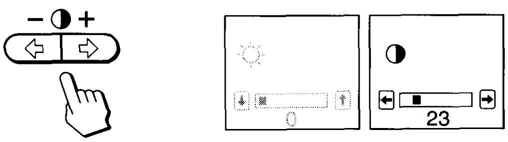
The ☼/◑ OSD disappears 3 seconds after you release the buttons.

To reset, press the →•← (reset) button while the OSD is on.

Adjusting the Picture Contrast

The adjustment data becomes the common setting for all input signals received.

1 Press the ◑⬅/➡ button.
The ☼/◑ OSD (On Screen Display) appears.



2 Press the ◑⬅/➡ buttons to adjust picture contrast.
⬅ . . . for less contrast
➡ . . . for more contrast

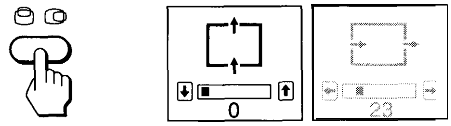
The ☼/◑ OSD disappears 3 seconds after you release the buttons.

To reset, press the →•← (reset) button while the OSD is on.

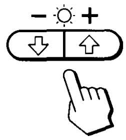
Adjusting the Picture Centering

The adjustment data becomes the unique setting for the input signal received.

1 Press the ⊖⊙ button.
The ⊖⊙ OSD (On Screen Display) appears.

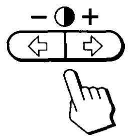


2 For vertical adjustment
Press the ☼⬇/⬆ buttons.



⬇ . . . to move down
⬆ . . . to move up

For horizontal adjustment
Press the ◑⬅/➡ buttons.



⬅ . . . to move left
➡ . . . to move right

To erase the ⊖⊙ OSD, press the ⊖⊙ button again.
The ⊖⊙ OSD automatically disappears 10 seconds after you release the buttons.

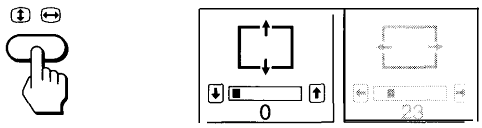
To reset, press the →•← (reset) button while the OSD is on.

EN

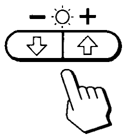
Adjusting the Picture Size

The adjustment data becomes the unique setting for the input signal received.

1 Press the ⇕⇔ button.
The GI ⇔ OSD (On Screen Display) appears.

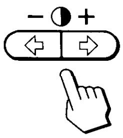


2 For vertical adjustment
Press the ☼ ⬇/⬆ buttons.



⬇ . . . to diminish
⬆ . . . to enlarge

For horizontal adjustment
Press the ◑ ⬅/➡ buttons.



⬅ . . . to diminish
➡ . . . to enlarge

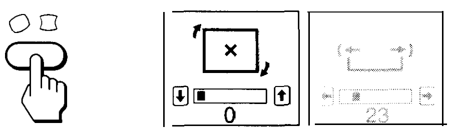
To erase the ⇕⇔ OSD, press the ⇕⇔ button again.
The ⇕⇔ OSD automatically disappears 10 seconds after you release the buttons.

To reset, press the →•← (reset) button while the OSD is on.

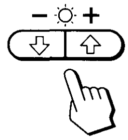
Adjusting the Picture Rotation

The adjustment data becomes the common setting for all input signals received.

1 Press the ◯◻ button.
The ◯◻ OSD (On Screen Display) appears.



2 Press the ☼ ⬇/⬆ buttons.



⬇ . . . to rotate counterclockwise
⬆ . . . to rotate clockwise

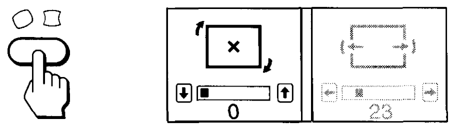
To erase the ◯◻ OSD, press the ◯◻ button again.
The ◯◻ OSD automatically disappears 10 seconds after you release the buttons.

To reset, press the →•← (reset) button while the OSD is on.

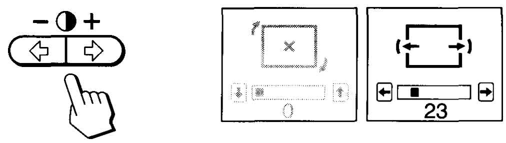
Adjusting the Pincushion

The adjustment data becomes the unique setting for the input signal received.

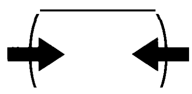
1 Press the ◯◻ button.
The ◯◻ OSD (On Screen Display) appears.



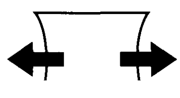
2 Press the ◑ ⬅/➡ buttons.



⬅ to diminish the picture sides



➡ to expand the picture sides



To erase the ◯◻ OSD, press the ◯◻ button again.
The ◯◻ OSD automatically disappears 10 seconds after you release the buttons.

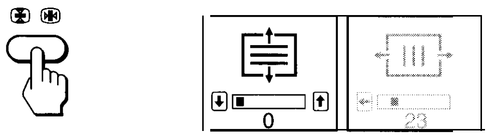
To reset, press the →•← (reset) button while the OSD is on.

Adjusting the Convergence

The adjustment data becomes the common setting for all input signals received.

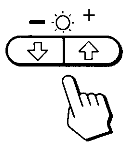
1 Press the ⊜⊞ button.

The ⊜⊞ OSD (On Screen Display) appears.



2 For vertical adjustment

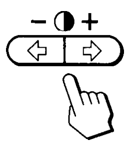
Press the ☼ ⬇/⬆ buttons.



⬇ . . . to move Red down and Blue up
⬆ . . . to move Red up and Blue down

For horizontal adjustment

Press the ◑ ⬅/➡ buttons.



⬅ . . . to move Red to the left and Blue to the right
➡ . . . to move Red to the right and Blue to the left

To erase the ⊜⊞ OSD, press the ⊜⊞ button again.
The ⊜⊞ OSD automatically disappears 10 seconds after you release the buttons.

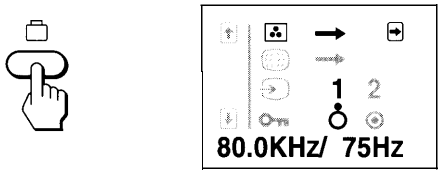
To reset, press the ➝⬸ (reset) button while the OSD is on.

Setting the Color Temperature

The selected color temperature becomes the common setting for all input signals received.

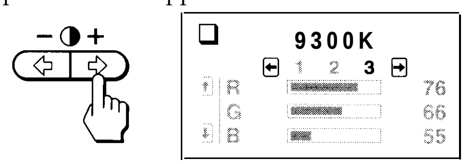
1 Press the ⌂ (option) button.

The option OSD (On Screen Display) appears.



2 Press the ◑ ➡ button.

The color temperature OSD appears.

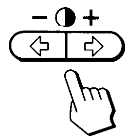


3 Adjust with the ◑ ⬅/➡ and ☼ ⬇/⬆ buttons.

There are three color temperature modes on the OSD. The factory settings are 1 : 5000K, 2 : 6500K, and 3 : 9300K.

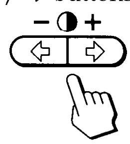
To select 5000K, 6500K or 9300K

Press ◑ ⬅/➡ buttons.

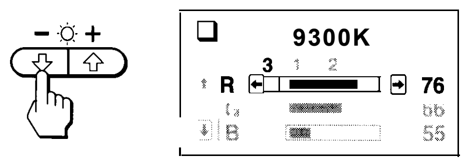


To obtain the desired color temperature

1 Press ◑ ⬅/➡ buttons to select the 1, 2 or 3 mode.



2 Press ☼ ⬇/⬆ buttons to select R, G or B.



3 Press ◑ ⬅/➡ buttons to adjust the color temperature.

The "5000K," "6500K" or "9300K" disappears.
The new color temperatures are memorized for each of the color modes -1, 2, 3.

To erase the OSD, press the ⌂ button again.
The OSD automatically disappears 30 seconds after you release the buttons.

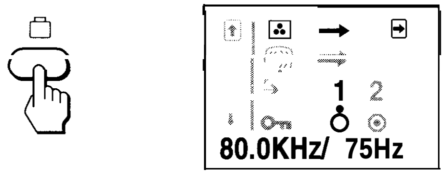
To reset, press the ➝⬸ (reset) button while the OSD is on.

EN

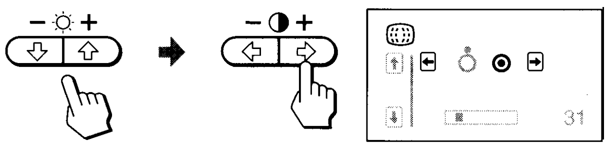
Canceling the Moire

The adjustment data becomes the unique setting for the input signal received.

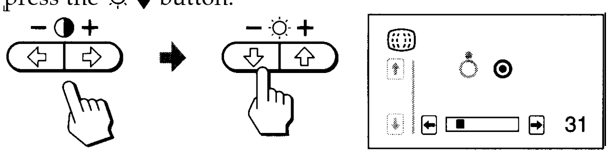
1 Press the □ (option) button.
The option OSD (On Screen Display) appears.



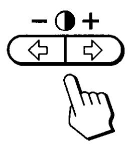
2 Press the ☼ ⬇/⬆ buttons to select ▦ and then press the ◑ ➡ button.



3 Press the ◑ ⬅/➡ buttons to select "◉" and then press the ☼ ⬇ button.



4 Press the ◑ ⬅/➡ buttons to adjust for minimum moire. Adjust the moire gradually from 0, and stop the adjustment as near to 0 as possible.



To erase the OSD, press the □ button again.
The OSD automatically disappears 30 seconds after you release the buttons.

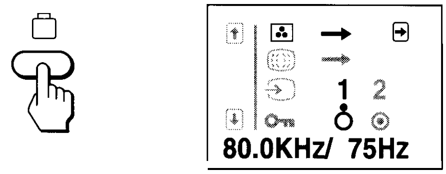
To reset, press the →•← (reset) button while the OSD is on.

Note
When the moire is cancelled, the picture may be fuzzy.

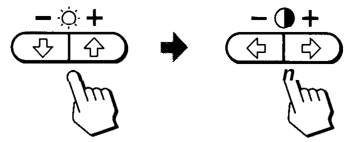
Switching the Input Connector

It is necessary to select the connector type (HD15/5 BNC) correctly according to the connection.

1 Press the □ (option) button.
The option OSD (On Screen Display) appears.



2 Press the ☼ ⬇/⬆ buttons to select ⇥ and then press the ◑ ⬅/➡ buttons to select "1" or "2."



"1" is for the Video input 1 connector (HD15) and "2" is for the Video input 2 connector (5 BNC).

If you select Video input 1 or 2 connector and that connector does not receive any signal, the unit automatically switches to the other connector.

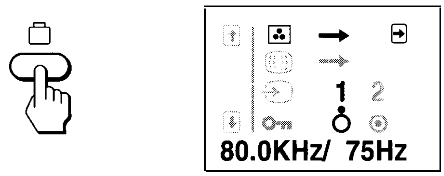
To erase the OSD, press the □ button again.
The OSD automatically disappears 30 seconds after you release the buttons.

To reset, press the →•← (reset) button while the OSD is on.

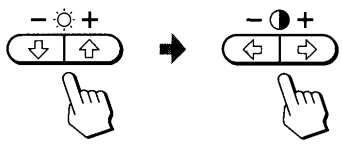
Locking the Controls

Locks the monitor so that the buttons, except the ⏻ power switch and ☐ (option) button, on the front panel cannot be operated (Lock mode). With this function, you can ensure that the display remains in the optimum condition even when the monitor is placed where the control buttons can be touched by unauthorized persons.
To release the Lock mode, set it to off ( ) position.
Normally keep this function set to off ( ) position.

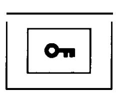
1 Press the ☐ (option) button.
The option OSD (On Screen Display) appears.



2 Press the ☼↓ button to select “ ” and then press the ◑←/→ buttons to select “⊙.”



Once you select “⊙,” you cannot select another item on the option OSD using the ☼↓/↑ buttons.
The control buttons on the front panel except the ⏻ power switch and ☐ (option) button do not work. If you press any button, the mark appears on the screen.



To Cancel the Control Lock

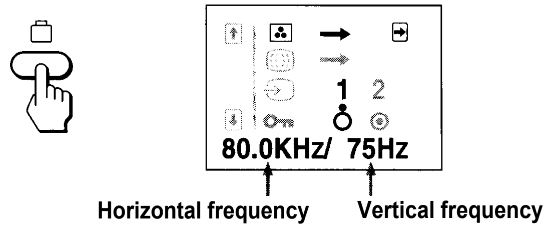
Press the ◑ ←/→ buttons to select “ .”

To erase the OSD, press the ☐ (option) button again.
The OSD automatically disappears 30 seconds after you release the buttons.

Checking the Signal Frequency

It is possible to check the current vertical and horizontal frequencies of the input signal received.

Press the ☐ (option) button.
The option OSD (On Screen Display) appears.



To erase the OSD, press the ☐ (option) button again.
The OSD automatically disappears 30 seconds after you release the buttons.

Resetting the Adjustment Data to Factory-preset Levels

EN

To reset an adjustment item

If you want to reset the color temperature

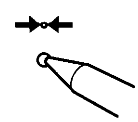
Select one of the three color temperature modes (see “Setting the Color Temperature” on page 7), and then press the →•← (reset) button while the OSD (On Screen Display) is on.

If you want to reset the other adjustment items

Press the button of the adjustment item you want to reset, and then press the →•← (reset) button while the OSD (On Screen Display) is on.

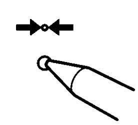
To reset the brightness, contrast, size, center and pincushion adjustment data at once (for the received signal)

Press the →•← (reset) button for 1 second when no OSD is shown.



To reset all adjustment data to factory-preset levels

Press and hold the →•← (reset) button for more than 2 seconds.
All adjustment data including the brightness and contrast are reset to factory-preset levels.



Power Saving Function

This monitor is capable of 3 states of reduced power consumption.

By sensing the absence of video signals and one or both sync signals coming from the host computer, it will reduce power consumption as follows.

	State	Power consumption	Required resumption time	POWER SAVING indicator	(power) indicator
1	Normal operation	≤ 140 W	—	off	green on
2	Standby (1st step of power saving)	≤ 100 W	approx. 3 sec.	orange on	green on
3	Suspend (2nd step of power saving)	≤ 15 w	approx. 3 sec.	orange on	green on
4	Active-off (3rd step of power saving)	≤ 5 W	approx. 10 sec.	orange on	off
5	Power-off	0 W	—	off	off

If both or any indicator is flashing, please cycle the power switch once. If they are still flashing, there is a potential monitor failure.
Please note which function indicator is lit, unplug the AC cord, and contact your service representative.

Power Saving Operation

- The H-sync is not present.

  ➡ The unit goes into standby state.

- The V-sync is not present.

  ➡ The unit goes into suspend state.

- Both the H-sync and V-sync are not present.

  ➡ The unit goes into active-off state.

The monitor requires a videocard or screen saver software which switches off one or both sync signals to activate the power saving function.

Caution: If the power switch is turned on without any video signal input, the Power Saving function automatically puts the monitor into the Active-off state and the POWER SAVING indicator lights up. Once the horizontal and vertical syncs are sensed, the monitor will automatically return to its Normal operation state.

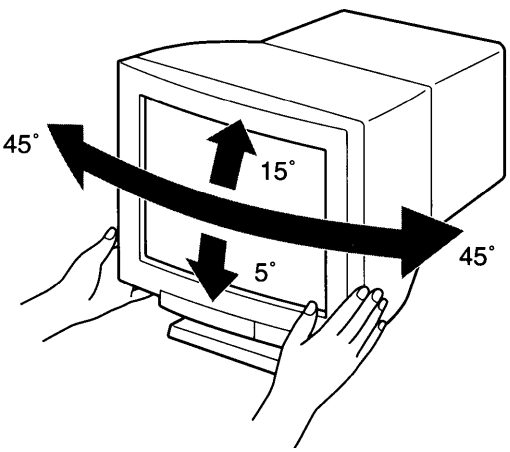
Plug and Play

This monitor complies with the DDC™1, DDC2B and DDC2AB which are the Display Data Channel (DDC) standards of VESA.
When a DDC1 host system is connected, the monitor synchronizes with the V. CLK in accordance with the VESA standards and outputs the EDID (Extended Display Identification Data) to the data line.
When a DDC2B or DDC2AB host system is connected, the monitor automatically switches to each communication.

DDC™ is a trademark of Video Electronics Standard Association.

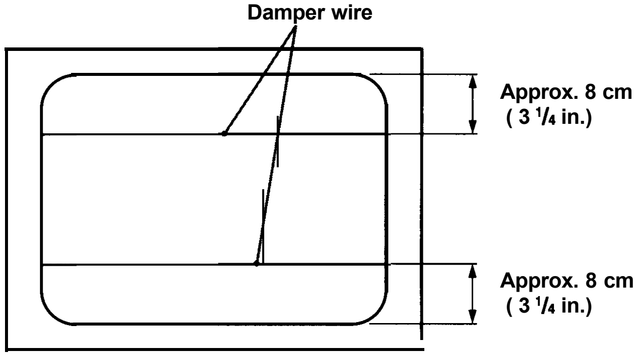
Use of the Tilt-Swivel

With the tilt-swivel, this unit can be adjusted to be viewed at your desired angle within 90″ horizontally and 20″ vertically.
To turn the unit vertically and horizontally, hold it at its bottom with both hands as illustrated below.



Damper Wire

Using a white background, very thin horizontal stripes on the screen are visible as shown below. These stripes are damper wires. These wires are attached to the aperture grille inside the Trinitron tube and are there to damp vibrations of the aperture grille in order to prevent them from influencing to the picture quality.

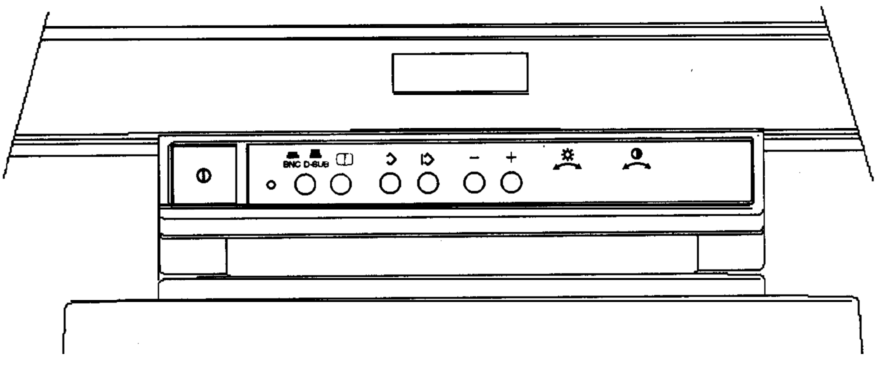


Specifications

Picture tube	0.25 mm aperture grille pitch 20 inches measured diagonally 90-degree deflection
Viewable image size	Approx. 387 x 292 mm (w/h) (15-1/4 x 11-1/2 inches) 19.0″ viewing image
Resolution	Horizontal: Max. 1600 dots Vertical: Max. 1200 lines (Recommended resolution: Horizontal: 1280 dots Vertical: 1024 lines)
Standard image area	Approx. 373 x 280 mm (w/h) (14 3/4 x 11 1/8 inches) or Approx. 350 x 280 mm (w/h) (13 7/8 x 11 1/8 inches)
Deflection frequency	Horizontal: 30 to 96 kHz Vertical: 48 to 160 Hz
AC input voltage/current	100 to 120 V, 50/60 Hz, 1.7 A 220 to 240 V, 50 – 60 Hz, 1.2 A
Power consumption	Max. 150 W
Dimensions	472 x 493.5 x 501 mm (w/h/d) (18 5/8 x 19 1/2 x 19 3/4 inches)
Mass	Approx. 30 kg (66 lb 2 oz)

Design and specifications are subject to change without notice.

CONTROL LOCATIONS AND FUNCTIONS



POWER
- Press this knob, the power indication LED lights and the power is ON
- Press this knob again, the power indication LED lights and the LED disappears and the power is OFF

DEGAUSS
- Eliminates the color impurity

D-SUB / BNC
- Switch to 'D-SUB' position, when the monitor is connected with a D-sub interface cable.
- Switch to 'BNC' position, when the monitor is connected with a BNC interface cable

CONTRAST
- Adjust contrast (the ratio between the brightness of the brighte-st and darkest parts of a picture)

BRIGHTNESS
- Adjust brightness (brillance of the screen)

SAVE
- Stores the result of adjustments (see Hardware OSD chapter)

'+' or '-'
- Selects or adjusts the functions (see Hardware OSD chapter)

3 **SELECT**
- Displays the On-Screen-Display (see Hardware OSD chapter)
- Accesses functions

HARDWARE OSD

The hardware On Screen Display (Hw OSD) will allow you to adjust the screen parameters and *select other special functions like displayed modes identification, password and power saving.*
The Hw OSD is organized on three levels:

First Level — **Second Level** — **Third Level**

- (H-Shift)
- (H-Size)
- (V-Shift)
- (V-Size)
- T. (Tilt)
- G. (Geometry)
  - (Pin/Barrel)
  - (Trapezoid)
  - (Parallelogram)
  - UBP (Unbalance Pin)
  - (Return)
- C. (Convergence)
  - (H-Convergence)
  - (V-Convergence)
  - (Return)
- RGB (True Color)
  - 9300
  - 6500
  - User (User Setting)
    - R (Red Color)
    - G (Green Color)
    - B (Blue Color)
    - (Return)
  - (Return)
- S. (Special Functions)
  - ID (Identification)
    - Alpha-numeric charac.
    - (Return)
  - P W (Password)
    - Numeric charac.
    - (Return)
  - P S (Power Saving)
  - (Return)

HOW TO START

The Hw OSD pictures will disappear after 8 seconds.

- When you switch on the monitor or you change the resolution, the Hw OSD will show the 'current video mode', e.g.:

1280 x 1024 90K/84 Hz

- if the 'current video mode' is one of the 12 **factory-preset timings, see** Appendix **4,** the Hw OSD will show the resolution and its working frequencies.

Note: This displayed mode can be disabled/enabled by pressing '+' and simultaneously for 2 seconds.

NEW MODE 90K/84 Hz

- if the current video-mode has not been saved yet, the Hw OSD will display the message 'NEW MODE'. You can either save the operating mode or continue working (you can save up to 18 new modes).

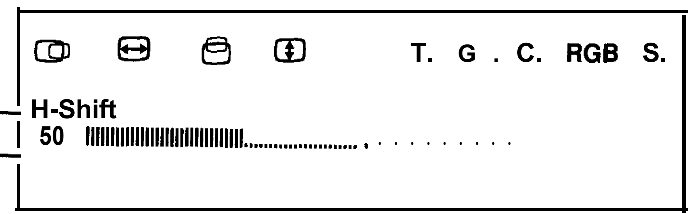
The user can give a personal name to all the video mode displayed (see **Identification** chapter)

SIZE POSITION AND SCREEN GEOMETRY

Size position and TILT adjustment
(Default to H-shift)

This chapter shows you how to adjust via Hw OSD the geomerical screen parameters:

1 Press key to access the first level, then the following window will appear on the screen:



Selected function symbol

'50 IIIIIIII' represents the magnitude of the current adjusted function'

Hw OSD **notes**

OSD will disappear automatically after about 30 seconds if no adjustment is performed (for the three levels).
- *Different colors of the OSD have different meanings :*
  WHITE-non-selected symbol
  YELLOW- selected symbol
  GREEN - confirmation of the selected symbol
- *Press '-' and '+' simultaneously to remove the OSD window*

2 Press '+' or '-' to select a function (e.g. select H-shift)

3 Press ◇ to confirm the selected function

4 Press '+' or '-' to adjust the magnitude of the change.

5 Press ⇧ key to quit the function mode or press ◇ key to store the adjustment (the screen will go back to mode display).

- For the next geometrical parameters (H-size, V-shift , V-size and Tilt) repeat the steps 2 - 5.

Screen geometry
(Default to 'Pin barrel')

The user may select ⇧ key to exit to first level by pressing ◇ key.

1 Press ◇ key to access the first level

2 Press '+' or '-' to select G. (=Geometry), the following window will appear on the screen:

UBP

Pin/Barrel
50 IIIIIIIIIIIIIIIIIIIIIIIIIIIII...............

3 Press ◇ to confirm the selected function.

4 *Now you* are on the'Second level'. Press '+' or '-' to select one of the available functions (Pin/Barrel or Trapezoid or Parallelogram or Unbalance pin) and than press ◇ to confirm the selected function.

5 Press '+' or '-' to adjust the magnitude of the change.

6 Press ⇧ key to quit the function mode or press ◇ key to store the adjustment (the screen will go back to mode display).

Convergence
(Default to 'H. Convergence')

1 Press ◇ key to access the first level

2 Press '+' or '-' to select C. (=Convergence), the following window will appear on the screen:

H-Convergence
50 IIIIIIIIIIIIIIIIIIIIIIIIIIIII...............

3 Press ◇ to confirm the selected function.

4 Now you are in the'Second level'. Press '+' or '-' to select **H-Convergence** (to remove convergence-error of vertical lines) or **V-Convergence** (to remove convergence error of horizontal lines) and than press ◇ to confirm the selected function.

5 Press '+' or '-' to adjust the magnitude of the change.

6 Press ⇧ key to quit the function mode or press ◇ key to store the adjustment (the screen will go back to mode display).

Recall factory preset **and delete user adjustment**

- Press ◇ and '-' simultaneously to recall factory preset settings and delete all the geometrical user adjustments.

WHITE COLOR TEMPERATURE

RGB 'True color' selection (Default to '9300°K')

1 Press ◇ key to access the first level

2 Press '+' or '-' to select RGB **True Color,** and press ◇ to confirm the selected function.
The following window will appear on the screen:

```
9300        6500        User        ⇧
9300.
```

3 Now you are in the 'Second level', press '+' or '-' to select the factory white color temperatures (9300 or 6500) or the adjustable color temperature **(User).**

If you select either 9300 or 6500 and press ◇ you go to paragraph 6; if you select **User** and press ◇ you go to the next paragraph.

4 Press ◇ key to access to the third level and press again '+' or '-' to select one color (R =Red or G=Green or B=Blue), the following window will appears on the screen:

```
R         G         B                ⇧
Red Color

50  IIIIIIIIIIIIIIIIIII...............
```

5 Press ◇ to confirm the selection, and than press '+' or '-' to adjust this color. If necessary repeat this step for the other colors.

6 Press ⇧ key to exit to select function mode or press ◇ key to store the adjustment (the screen will go back to mode display).

'Special' functions
(Default to I.D.)

1 Press ◇ key to access the first level

2 Press '+' or '-' to select S. (=Special), and than press ◇ to confirm the selection and enter the second level.
The following window will appears on the screen:

```
                                     ⇧
ID         PW         PS
IDENTIFICATION
```


3 Press '+' or '-' to select one of the available functions:

- ID = IDENTIFICATION. This function allow you to give a personal names to all the video mode

- PW = PASSWORD. This function allow you to set in a password and than prevent monitor changes from unauthorized users.

- PS = POWER SAVING. This function allow you to enable/disable the power management.

ID : Identification

1 Press ◇ to confirm the selection. Now the following window appears on the screen

```
          1152x900                  row 1
0123456789ABCDEFGHIJ                row 2
KLMNOPQRSTUVWXYZ_/.                 row 3
```

2 On the 'row 1' the OSD will displays the latest mode name, if any, and on the other rows the available characters.

Press ◇ key to access to the alpha-numeric characters on second and third row, than press '+' or '-' keys to select the first character of your personal name.
Confirm the selection by pressing ◇ key, then the selected character will be shown on row 1 and cursor moves to the next position. Repeat this step for all the other characters.

3 Press ⇧ key to quit the function mode or press ◇ key to store the adjustment (the screen will go back to mode display).

PW: Password

A) To set & confirm password

If you want to clear the password after a password is set & confirmed, please enter the old password and than the characters '000' according to the sequence described in this section.

Cautions:

* ***Make sure that you memorize the password & enter the right one***
* ***please use a password only if necessary: there could be the risk to forget it. In this case p/ease contact Service.***

1 Press ▷ to confirm the selection. Now the following window appears on the screen

Setting Password	row 1
New Password: 000 ⇧	row 2
(0123456789 ⇧)	row 3

2 Press ▷ key to access to the numeric characters on third row, than press '+' or '-' keys to select the first character of your password. Repeat this step for all the other characters.

3 Press ⇧ and than ▷ keys to quit the password window or press ⇨ key to store the password. In this case the next window will appear on the screen:

Setting Password
Confirmation : 000 **⇧**
(0123456789 ⇧)

4 Key in the password again to confirm and press ⇧ to exit to the second level.

B) Password check

If a password has been set up by user, whenever the user wants to save the new setting and press ⇨ key at any stage, the OSD will show:

Check Password
Old Password: 000
(0123456789 ⇧) ⇧

1 Key in the right password by the same procedure described on chapter *A)*, and press ⇨.

- *If the password is wrong, then the following window will appear on the screen:*

Password Error

- *Press any one of ▷,'-','+', ⇨ keys to exit to previous password-checking window.*

C) To change the password

If the password ***has been set,*** *whenever the user selects 'P. W. ' the OSD will show :*

Modify Password
Old Password: 000 ⇧
(0123456789 ⇧)

- If the old password is wrong-given, the OSD will show the message 'Password Error', then the user can exit to previous screen by pressing one of the following keys ▷ ,'-', '+', ⇨
- After the old password has been key in correctly, then the OSD will display the previous 'Setting Password' screen to modify the password.
- Press ⇧ key to quit the password window.

PS : Power saving

The monitor has a power saving function. When the monitor does not receive horizontal and/or vertical ***synchronisms,*** *after a certain period of time, it will go into orie of the following power saving modes.*

1 Press ◇ to confirm the selection. Now the following window appears on the screen:

ID PW PS
Power Saving
Now is ON

2 Press ◇ key to select the function 'ON' (enable') or 'OFF (disable) and than press ▷ to confirm the selection.

Power Management Definition

VESA DPMS				POWER SAVING STATES	
MODE	VIDEO	H-SYNC	V-SYNC	POWER USED	POWER SAVING (%)
ON	Active	Yes	Yes	100 Watts	0 %
Stand-by	Blank	No	Yes	< 15 Watts	> 86 %
Suspend	Blank	Yes	No	< 15 Watts	> 86 %
OFF	Blank	No	No	< 5 Watts	> 95 %

Note: The video will be blanked and the power LED color will change into following colors whenever the monitor goes into power management status.

Green light - 'Normal' operation
Yellow light - 'Stand by' or 'Suspend' mode
Amber light - 'OFF' mode

This monitor is Energy Star compliant when used with a computer equipped with DPMS.

The manufacturer is a partner in the EPA's Energy Star Computer Program.

CLEANING AND MAINTENANCE

Please remember that if you take a good care of your monitor it will give you many years of trouble free usage.

Important!

When cleaning your monitor:

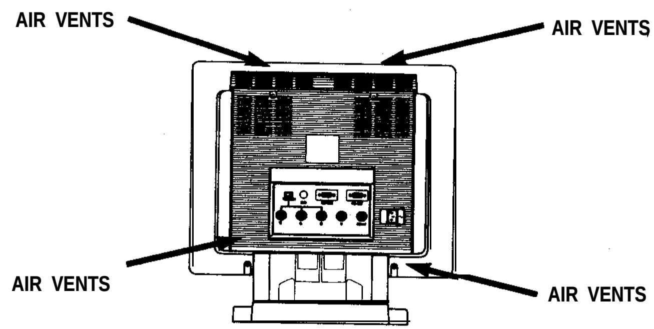
- always switch the monitor off
- always use a damp and NOT WET soft cloth
- clean the screen using a household glass cleaner sprayed on the soft cloth and then wipe the screen
- never use aggressive cleaning agents on the surface of the monitor. They might discolor the cabinet and affect the anti-glare treatment on your screen

SAFETY SUGGESTION

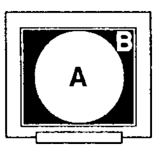
Remember that your monitor is a device with extremely complex and delicate circuitry that can be damaged by rough treatment.

Important !

- Do not open the monitor. Repairs must be carried out by qualified service personelle.
- Do not leave your monitor exposed to sunlight. Intensive heat can damage the internal circuitry of the monitor.
- Do not place cups, glasses or anything else containing liquid on or beside the monitor
- Do not obstruct the 'air vents' (see picture below) with paper, books, or other material. The vents are essential to cool the monitor.



The following specification are subject to change wilhout notice. Specification are guaranted for the pre-set timings at the nominal picture size.

	TECHNICAL SPECIFICATIONS
CRT	20 inch (18.81 " V) - 90° Deflection, neck diameter 29.1 mm Tinted screen glass, Black Matrix, Trinitron® cathode ray tube 0,26 mm aperture grille pitch - Phosphor P22 - γ = 2.15 Phosphor CIE coordinates: RED x=0.625 y=0.340 GREEN x=0.280 y=0.595 BLUE x=0.155 y=0.070 AR treatment (Anti-Reflection) with Anti-Static treatment 39 % light transmission (approx.) maximum useful screen sizes (mm) = 387 (H) x 291 (V)
POWER SUPPLY	Universal Power Supply 90 to 270 Vac.
POWER CONSUPTION	125 Watts typ. (140 Watts max)
INPUT SIGNALS	VIDEO: RGB analog.,0.714 Vpp typ. (impedance = 75Ω) BNC inputs impedance=75Ω or high impedance, selection by external switch D-SUB inputs impedance 75Ω SYNC: composite on-green or separate composite TTL level or separate H and V sync TTL level
SCANNING FREQUENCIES	HORIZ.RANGE: 30 to 95 kHz (continuous/autoscan) VERT. RANGE: 50 to 160 Hz (continuous/autoscan)
PIXEL CLOCK	157.5 MHz
RESOLUTION	up to 1280 x 1024 at 85Hz
MISCONVERGENCE	 0.3 mm max. in zone A (285 mm circle) and 0.4 mm max in zone B
GEOMETRIC DISTORSIONS	Pincushion, Trapezoidal, parallelogram, rotation and other various distortions must fall within a perimetral rail of 2 mm.
WHITE COLOR BALANCE (color temperature)	C.T.1 = 9300°K+8 mpcd - CIE coordinates X=0.283+/- 0.02 Y=0.298 +/-0.02 C.T.2 = 6500°K+7 mpcd - CIE coordinates X=0.313+/- 0.02 Y=0.329 +/-0.02 C.T.3 = user adjustable
BRIGHTNESS UNIFORMITY	> 75 %
COLOR UNIFORMITY	< 0.015 units (X and Y)

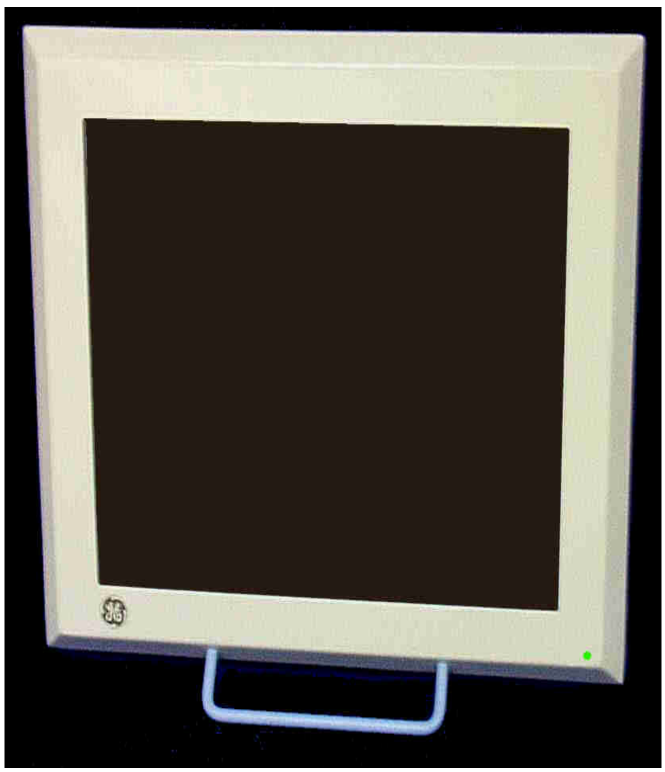
EMI/RFI	according to FCC-15 class B EN55022/ CISPR22 class B, EN80082-1, Electrostatic/magnetic fields and AC electrical in compliance with MPRII standards
SAFETY REGULATIONS	IEC 950/EN 60950, UL1950, CAN/CSAC22.2 N°950-93, DHHS CFR rule 21 subch.J, Marking: TUEV-GS, UL, UL-C, FCC, VCCI, CE.
ERGONOMICS	Ergonomics performance complies with ZH 1/618 and ISO9241-3
	ENVIRONMENTAL CONDITIONS
OPERATING	AMBIENT TEMP. : 0°C (32°F) to + 35°C (95°F)
STORAGE	AMBIENTTEMP.: -25°C (-13°F) to + 70°C (158°F)
HUMIDITY	10% to 90% R.H. (non condensing)
RELIABILITY	
MTBF	50,000 hours (excluding CRT)
	CONNECTORS (rear side)
INPUT CONN. SIGNALS	15 pin mini D-SUB female and 5 BNC' s
RS232 INPUT	standard D-SUB (only for Service personnel)
AC POWER	EN60320 C14 (EUROPE)
	EXTERNAL CONTROLS
REAR SIDE	Loop through switch (75 Ω)
FRONT SIDE	Power on-off - Brightness - Contrast - Manual degauss - BNC-D-sub selector - Digital control button with on screen display - Save,+,- and select keys for H-shift, H-size, V-shift, V-size, Pin/Barrel, Trapezoid, Pin Balance, Parallelogram, Convergence, Color temp. Color Red, Color Green, Color Blue,Tilt, mode identification and recall.
	CABLES
SIGNAL CABLE and POWER CORD	

ALLUS LCD MODEL ATC-5720-GE

1.0 ATC - 5720-GE

FLAT PANEL MONITOR

USER'S GUIDE



ATC – 5720 – GE
USER'S GUIDE

TABLE OF CONTENTS

1.0 INTRODUCTION

The ATC – 5720 – GE is a 20-inch, full color, SXGA LCD flat panel monitor. This monitor has been designed by ALLUS Technology Corporation for General Electric Medical Systems and is a certified Medical Device.

The intended use of the ATC – 5720 – GE USER'S GUIDE is to provide information on the installation, operation, troubleshooting, care and maintenance of the unit.

2.0 FEATURES

The ATC – 5720 – GE features include:

- Latest LCD flat screen technology with wide viewing angle
- 20 inch viewable display area
- 1280 X 1024 (SXGA) resolution
- RGB, sync on green full color monitor
- Compliance with FCC, UL, CSA, CE, Medical Device Directives, and EMI Compatibility standards. *

*SEE APPENDICES for detailed listing.

3.0 INSTALLATION

The ATC – 5720 - GE to be mounted, using the four mounting holes on the back panel.

To install:
(Refer to figure 1)

1. Mount monitor using the VESA mounting pattern on the rear of the unit using 10 – 32 screws. *

2. Connect the RGB video input to the 3 BNC connectors on the rear of the display. Red cable to Red, Green cable to Green, and Blue cable to Blue.

3. Connect the power to the monitor. Using the supplied IEC 320 power cord. (The power supply is autosensing, 90V to 240V acceptable.)

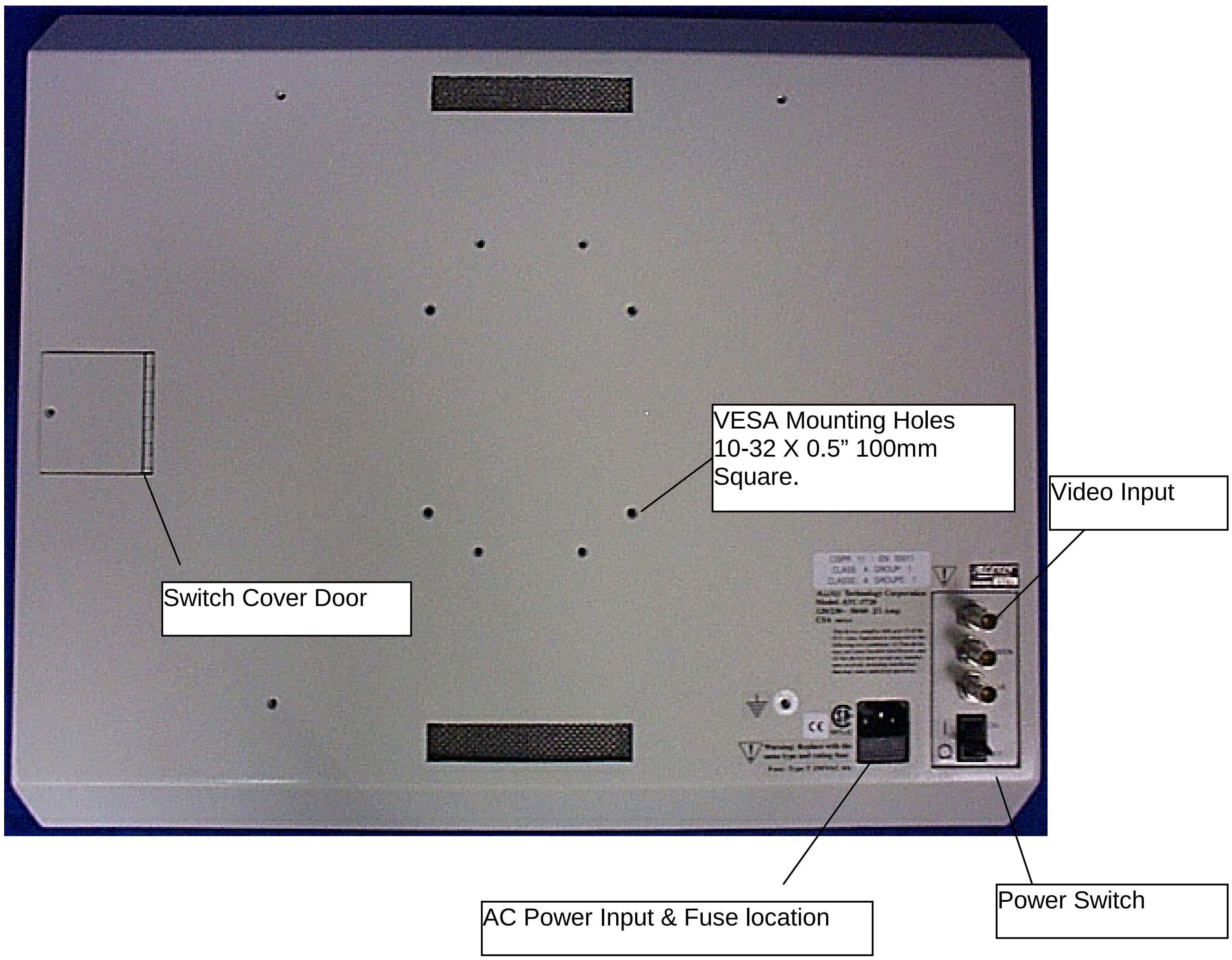
4. Turn on power. Using the switch on the right rear of display.

5. Using test pattern video, adjust the position, brightness, and clarity of the image using the control buttons on the back panel. (See Section 4.0 for adjustment instructions.)

6. The monitor position may be changed using the blue handle located below the screen, on the front of the unit.

*See Appendices for mounting hole specifications.

Figure 1.


VESA Mounting Holes
10-32 X 0.5” 100mm
Square.
Video Input
Switch Cover Door
AC Power Input & Fuse location
Power Switch

4.0 OPERATION

The ATC - 5720 – GE requires no operator intervention once the initial adjustments are made. However, if the monitor is moved to another system or video source, the controls may need to be adjusted to re - center the image on the screen.

The control buttons are located behind an access door found on the left back of the unit. The door is held in place with a phillips type screw. The access door should always be closed and secured after making adjustments.

NOTE: THERE ARE TWO VERSIONS OF CONTROL BUTTONS. THE INSTRUCTIONS FOR BOTH SETS ARE CONTAINED WITHIN THIS MANUAL.

Version one consists of eight push buttons, with the button labels located inside the access door. (See figure 2)

Version two consists of three push buttons and 5 LED indicators. The labels are on the access door as well as adjacent to the push buttons and LED indicators.
(See figure 3)

For both versions of controls the following adjustments apply:

BRIGHTNESS : This function allows the user to vary the backlight brightness of the LCD from full brightness to OFF. There are 256 steps from OFF to full brightness.

HORIZONTAL SIZE : This adjustment has the effect of expanding or contracting the image horizontally. It is used to “fit” the image on the screen left to right.

4.0 OPERATION, CONTINUED

CLOCK ADJUST : This is the focus adjustment.

VERTICAL POSITION : This adjustment moves the entire image up and down.

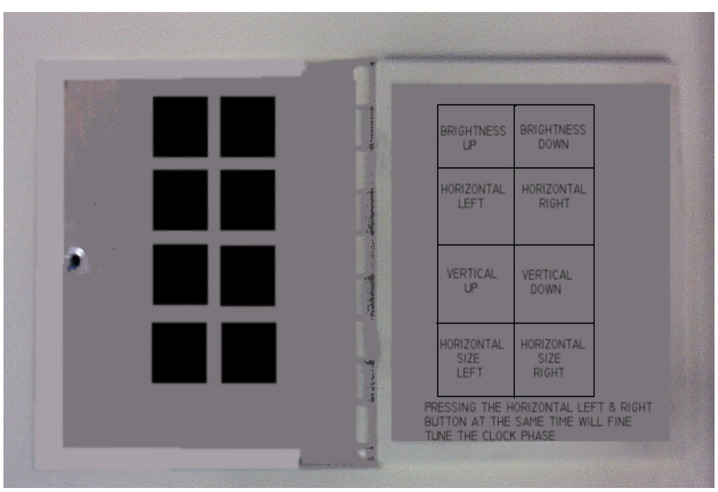
HORIZONTAL POSITION : This adjustment moves the entire image left and right.

For best results in adjusting the controls, the video test image should use the entire screen. This is 1280 X 1024 pixels of information. (Most computers have a test pattern that occupies the entire screen. If not, the video input may not be 1280 X 1024.)

For best results in adjusting the CLOCK ADJUST (focus), put a test pattern on the screen of single width vertical lines going across the entire screen. Step the CLOCK ADJUST until each vertical line is clear and clean.

EIGHT BUTTON ADJUST

Figure 2



4.0 OPERATION, CONTINUED

ADJUSTING THE CONTROLS FOR FIGURE 2:

Open the panel by removing the phillips type screw. The control buttons are:

Brightness UP
Brightness DOWN
Horizontal Position LEFT
Horizontal Position RIGHT
Vertical Position UP
Vertical Position DOWN
Horizontal Size LEFT
Horizontal Size RIGHT

To adjust the image on the screen, first adjust the vertical position. Then, adjust the horizontal position so that the left side of the image lines up on the left side of the screen. Next, adjust the horizontal size until the right edge of the image line up on the right side of the screen. It may be necessary to adjust the horizontal position and size a few times as changing one will affect the other.

When the image is positioned on the screen then adjust the focus, if necessary. To make this adjustment, press the Horizontal Position LEFT and Horizontal RIGHT buttons simultaneously. Focusing the image may not be needed if the horizontal size adjustment is made correctly.

4.0 OPERATION, CONTINUED

THREE BUTTON ADJUST

Press the SELECT button until the LED is active for the desired adjustment. Pressing the UP / DOWN keys will make the adjustment. Holding the UP / DOWN keys will cause them to repeat.

To adjust the image on the screen, first adjust the vertical position. Then, adjust the horizontal position so that the left side of the image lines up on the left side of the screen. Next, adjust the horizontal size until the right edge of the image line up on the right side of the screen. It may be necessary to adjust the horizontal position and size a few times as changing one will affect the other.

When the image is positioned on the screen then adjust the focus, if necessary. Press the SELECT button until the CLOCK ADJUST LED comes on. Then, using the UP / DOWN buttons, focus the image. Focusing the image may not be needed if the horizontal size adjustment is made correctly.

5.0 CARE AND MAINTENANCE

The ATC – 5720 – GE requires very little routine maintenance. If cleaning is needed use diluted household bleach, or a mild detergent, and a damp cloth.

If it becomes necessary to store the ATC – 5720 – GE monitor it is imperative the storage temperature not drop below –20 ° C or go above +60 ° C as temperature extremes may cause permanent damage to the LCD display.

The light source the ATC – 5270 - GE consists of 10 fluorescent tubes (CCFT tubes). The tubes have an average life span of 20,000 to 30,000 hours under normal use. To maximize the tube life, turn the unit off when not in use.

6.0 TROUBLESHOOTING

Below is a listing of the most common problems found during normal operation of the ATC – 5720 – GE monitor.

PROBLEM	CAUSE	ACTION
Screen is black, but front LED is on	Incoming video is black	Check Computer for screen saver
Screen is black, and LED on front is blinking	No Video	Check that the Host Computer is on. Check that the Video cables are connected correctly.
Screen is black, and LED on front is not on or blinking	Power or sync problem	Check AC power input Be sure power switch is in the ON position Check fuse (see 3.0 for location) Verify GREEN video is plugged into green connector
Colors are wrong	Blue and red are switched	Swap red and blue, so that the red is plugged into the red connector, and the blue into the blue connector.
Image will not size	Incoming video is not correct	Verify that the computer is outputting 1280 X 1024
Sparkling pixels or fuzzy image	Clock phase is not correct	Adjust clock phase
Monitor is dim		Adjust brightness
Vertical light and dark pattern	Horizontal size is not correct	Adjust horizontal size
Dark area on screen	One or more light tubes maybe burned out	Replace tubes

7.0 APPENDICES

SPECIFICATIONS

ENVIRONMENTAL

OPERATING TEMPERATURE	0° C TO +35° C
STORAGE TEMPERATURE	- 20° C TO +60° C

Note: At -40°C the liquid crystal is frozen. There is a possibility that the glass and polarizer will crack, resulting in Permanent damage. Also operating the CCFT tubes at low temperatures shortens their life and causes the tubes to be dim.

HUMIDITY, OPERATING	20% TO 80%, Non - condensing
HUMIDITY, NON-OPERATING	5% TO 95% Non - condensing
ALTITUDE, OPERATING	4570m (15,000 ft)
ALTITUDE, NON-OPERATING	15,200m (50,000ft)

Note: High altitudes may affect the adhesives used with the LCD module.

VIBRATION	5 – 100 Hz with 1.2G @ 1 min/cycle, 10 times in each Direction (X, Y, Z)

7.0 APPENDICES, CONTINUED

SPECIFICATIONS

ENVIRONMENTAL

SHOCK — 30G, 11msec, 10 times in each Direction (X, Y, Z)

MEAN TIME TO FAILURE (MTTF)

Controller Board	100,000 HRS.
Internal AC Power Supply	100,000 HRS.
LCD Module	31,000 HRS.

(Tested @ 25 °C for 8 hours ON and 16 hours off)

LCD Backlight	17,500 HRS.

(Tested @ 25 °C with continuous ON at full brightness, failure to first CCFT tube.)

5,800 HRS.

(Tested @ 25 °C switching 25 minutes ON and 5 minutes OFF until brightness goes to half brightness)

5,800 HRS.

(Tested @ 60 °C with continuous ON and full brightness to failure of first CCFT tube.)

Notes on MTTF:

1. Manufacturer experience indicates better reliability than the above numbers show.
2. The enclosure is designed to allow maximum conducted airflow while maintaining EMI/RFI emissions. There is more than1” to 1.5” of airflow area between the back of the LCD module and the rear panel of enclosure. The temperature rise inside the unit is less than 3 °C over ambient.
3. The least reliable component in the unit is the CCFT tube. There are 10 tubes per unit. The MTTF is based on the failure of the first tube. This does not render the monitor unusable. Rather, a small dark area will appear on the screen.

7.0 APPENDICES, CONTINUED

SPECIFICATIONS

MONITOR ATTRIBUTES

OVERALL DIMENSIONS	20.34” X 17.97” X 2.34”
DISPLAY AREA	15.72” X 12.57” (20.1 “ DIAGONAL)
DISPLAY TECHNOLOGY	a-si TFT active Matrix
DOT PITCH	0.104mm X 0.312mm
RESOLUTION	1280 X 1024 pixels
BRIGHTNESS	150 cd/m ^{2} (typ)
CONTRAST RATIO	220: 1 (typ)
RESPONSE TIME	130 msec (max) (Black to white)
NUMBER OF COLORS	Analog RGB (> 16M colors)
VIEWING ANGLE	Rated at contrast Ratio > 1:10 80 ° left to right 80 ° up and down

7.0 APPENDICES, CONTINUED

SPECIFICATIONS

MONITOR ATTRIBUTES

WEIGHT	16.75 lbs.
ENCLOSURE MATERIAL	0.090” aluminum
MOUNTING	10 – 32 threaded 0.5” deep blind PEM inserts.

Note: Mounting holes conform To VESA standard mounting for flat panel displays. Additional holes are provided for non-standard mounts.

REAR PANEL CONNECTORS	3 – BNC type 1 Red 1 Green 1 Blue 1 AC input power Connector
POWER REQUIREMENTS	90 - 240 VAC, 50 – 60 Hz, Autoswitching 90 – 120 VAC, @ 1A max 220 – 240 VAC.@ .5A max

Note: Power supply is UL approved. Power input connector conforms to IEC 320 Standard, with internal 4A Fuse.

7.0 APPENDICES, CONTINUED

SPECIFICATIONS

MONITOR ATTRIBUTES

POWER SWITCH	A toggle type power switch is located on the back of the unit next to the power input connector.
LENS COVER	Acrylic material, which is scratching resistant under normal use. It is resistance to bleach and is sealed to the enclosure using 3M VHB tape.
HANDLE	position adjustments handle is located on the lower front center of the monitor.
COOLING	Convection cooling using an air inlet on the back of the unit at the lower center and an air outlet at the upper center of the unit. The air inlet and outlet are covered with an EMC filter for emission compliance. There is no fan.

7.0 APPENDICES, CONTINUED

SPECIFICATIONS

VIDEO INTERFACE

Input Signal	RS – 343A, analog RGB (sync on green). 0.7V p-p, 75 ohm termination. Video is input on 3 separate BNC coax connectors for RED, GREEN/SYNC, BLUE.
Horizontal Rate	30 to 85 kHz
Vertical Rate	50 to 160 Hz
Video Mode	1280 X 1024

7.0 APPENDICES, CONTINUED

REGULATORY COMPLAINCE

FCC Class A

"Note: This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense."

CAUTION: CHANGES OR MODIFICATIONS NOT EXPRESSLY APPROVED BY THE MANUFACTURER OF THIS DEVICE CAN CAUSE VIOLATION OF COMPLAINCE AND VOID THE FCC RATING OF THIS DEVICE.

7.0 APPENDICES, CONTINUED

REGULATORY COMPLAINCE

The ATC - 5720 – GE holds the following regulatory ratings, as indicated by the labeling on the right lower quadrant of the rear panel:

UL
CSA
EN60950
CE Mark

The ATC - 5720 – GE is certified to the following Medical Device Directives:

IEC601.1
UL2601
CSA601-1

The ATC – 5720 – GE meets the following Electromagnetic Compatibility criteria:

Directive 93/42/eec for Europe
DR-4 for Australia
ICES-001 for Canada
MHW for Japan

The ATC – 5720 – GE adheres to Good Manufacturing Practices, as set forth by the FDA Quality Systems Requirements.

7.0 APPENDICES, CONTINUED

NOTICES

ALLUS TECHNOLOGY CORPORATION SHALL NOT BE LIABLE FOR TECHNICAL OR EDITORIAL ERRORS OR OMMISIONS CONTAINED HEREIN; NOR FOR INCIDENTAL OR CONSEQUENTIAL DAMAGES RESULTING FROM THE FURNISHING, PERFORMANCE, OR USE OF THIS MATERIAL.

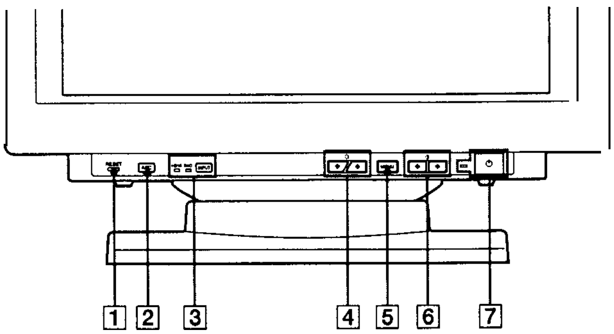
USER'S GUIDE

ATC – 5720 – GE
First Edition (October 1998)

Identifying Parts and Controls

See the pages in parentheses for further details.
GDM500PS is used for illustration purposes throughout this manual.

Front



1 RESET (reset) button (page 17)
Resets the adjustments to the factory settings.

2 ASC (auto sizing and centering) button (page 7)
Automatically adjusts the size and centering of the images.

3 INPUT (input) button and HD15/BNC indicators (page 8)
Selects the HD15 or 5BNC video input signal. Each time you press this button, the input signal and corresponding indicator alternate.

4 ☼ (brightness) (↓/↑) buttons (pages 8 – 17)
Adjust the picture brightness.

Function as the (↓/↑) buttons when adjusting other items.

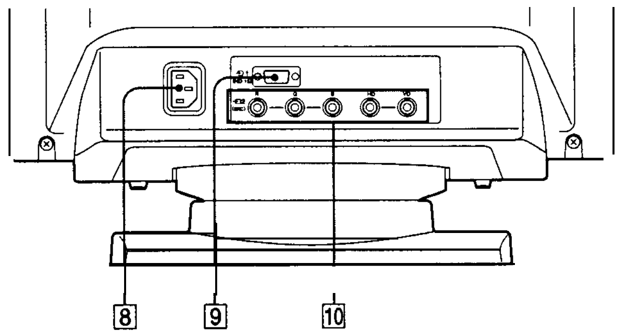
5 MENU (menu) button (pages 8 -17)
Displays the MENU OSD.

6 ◑ (contrast) (←/→) buttons (pages 8 – 17, 22)
Adjust the contrast.

Function as the (←/→) buttons when adjusting other items.

7 ⏻ (power) switch and indicator (pages 19, 22)
Turns the monitor on or off.
The indicator lights up in green when the monitor is turned on, and either flashes in green and orange or lights up in orange when the monitor is in power saving mode.

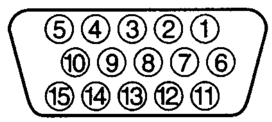
Rear



8 AC IN connector
Provides AC power to the monitor.

EN

9 Video input 1 connector (HD15)
Inputs RGB video signals (0.700 Vp-p, positive) and SYNC signals.



Pin No.	Signal	Pin No.	Signal
1	Red	8	Blue Ground
2	Green (Composite Sync on Green)	9	DDC + 5V*
		10	Ground
		11	ID (Ground)
3	Blue	12	Bi-Directional Data (SDA)*
4	ID (Ground)		
5	DDC Ground*	13	H.Sync
6	Red Ground	14	v. sync
7	Green Ground	15	Data Clock(SCL)*

* Display Data Channel (DDC) Standard of VESA

10 Video input 2 connector (5 BNC)
Inputs RGB video signals (0.700 Vp-p, positive) and SYNC signals.

Setup

Before using this monitor, check that the following items are included in your carton:

- Monitor (1)
- Power cord (1)
- HD15 video signal cable (1)
- Macintosh adapter (1)
- Windows Monitor Information Disk (1)
- TCO' 95 Eco-document (1) (GDM500PS only)
- Warranty card (1)
- These operating instructions (1)

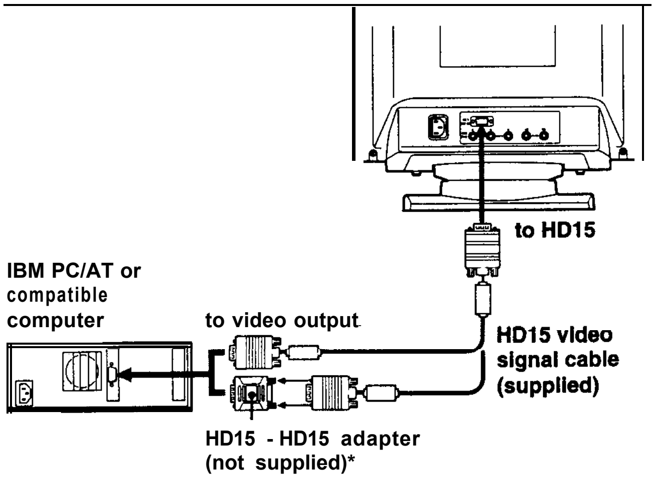
This monitor works with any IBM or compatible system equipped with VGA or greater graphics capability. Although this monitor works with other platforms running at horizontal frequencies between 30 and 94 kHz (GDM-400PS), 30 and 107 kHz (GDM500PS), including Macintosh and Power Macintosh systems, a cable adapter is required. Please consult your dealer for advice on which adapter is suitable for your needs.

Step 1: Connect the monitor to the computer

With the computer switched off, connect the video signal cable to the monitor using the supplied HD15 video signal cable.

- If you are using an IBM PC/AT or compatible computer, refer to the section below.
- If you are using a Macintosh or compatible computer, refer to the following section, "Connecting to a Macintosh or compatible computer."
- If you want to use the 5 BNC connectors, refer to the section, "Connecting to the 5 BNC connectors."

Connecting to an IBM PC/AT or compatible computer

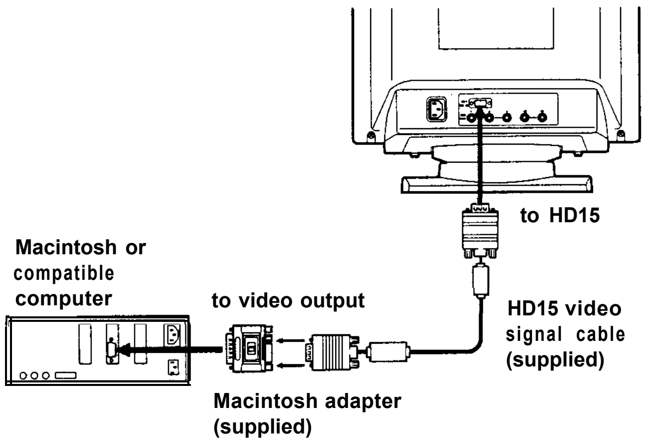


* The HD15 - HD15 adapter may be needed for some models.

If your PC system is not compatible with DDC2AB and DDC2B+

This monitor uses the No. 9 pin in the video signal connector for DDC2AB and DDC2B+ compatibility.
Some PC systems which are not compatible with either DDC2AB or DDC2B+ may not accept the No. 9 pin. If you are not sure whether your PC system accepts the No. 9 pin or not, use the HD15 (Female) - HD15 (Male without the No. 9 pin) adapter (not supplied). Make sure the male side (without the No. 9 pin) is connected to the computer.

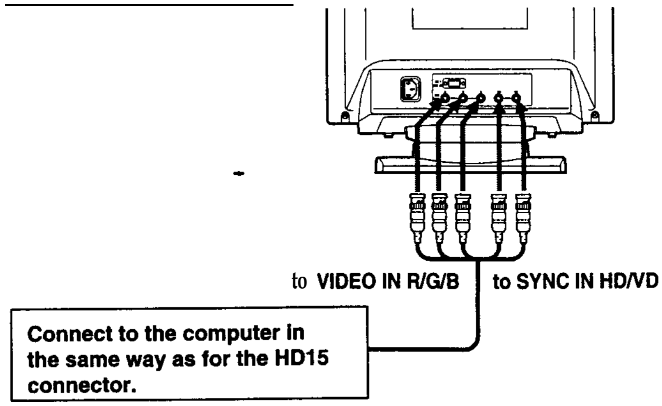
Connecting to a Macintosh or compatible computer



About the supplied Macintosh adapter

The supplied Macintosh adapter is compatible with Macintosh LC, Performa, Quadra, Power Macintosh and Power Macintosh G3 series computers.
Macintosh II series and some older versions of Power Book models may need an adapter with micro switches (not supplied).

Connecting to the 5 BNC connectors



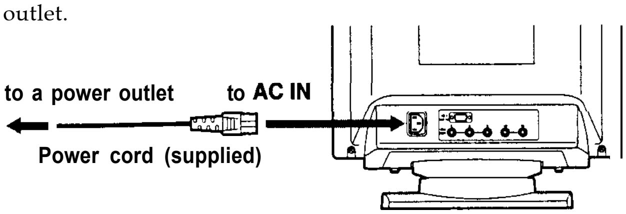
To connect the 5 BNC connectors, use the SMF-400 video signal cable (sold separately). Connect the cables from left to right in the following order: Red-Green-Blue-HD-VD.

Notes

- Do not short the pins of the video signal cable.
- The DDC standard does not apply to the 5 BNC connectors. If you use the DDC standard, connect the HD15 connector to the computer with the supplied video signal cable.

Step 2: Connect the power cord

With the monitor switched off, connect one end of the power cord to the monitor and the other end to a power outlet.



Step 3: Turn on the monitor and computer

The installation of your monitor is complete.

Note

If "OUT OF SCAN RANGE" or "NO INPUT SIGNAL" appears on the screen, see "Warning Messages" on page 20.

For customers using Windows 95/98

To maximize the potential of your monitor, install the new model information file from the supplied Windows Monitor Information Disk onto your PC.

This monitor complies with the "VESA DDC" Plug & Play standard. If your PC/graphics board complies with DDC, select "Plug & Play Monitor (VESA DDC)" or this monitor's model name as the monitor type in the "Control Panel" of Windows 95/98.

If your PC/graphics board has difficulty communicating with this monitor, load the Windows Monitor Information Disk and select this monitor's model name as the monitor type.

For customers using Windows NT4.0

Monitor setup in Windows NT4.0 is different from Windows 95/98 and does not involve the selection of monitor type. Refer to the Windows NT4.0 instruction manual for further details on adjusting the resolution, refresh rate, and number of colors.

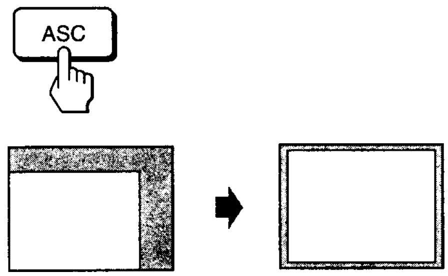
Automatically Adjusting the Size and Centering of the Picture

By pressing the auto sizing and centering (ASC) button, the size and centering of the picture are automatically adjusted to fit the screen.

1 **Turn on the monitor and computer.**

2 **Press the ASC button.**

The picture is adjusted to fit the center of the screen.



Notes

- This function is intended for use with a computer running Windows or similar graphic user interface software that provides a full-screen picture. It may not work properly if the background color is dark or if the input picture does not fill the screen to the edges (such as an MS-DOS prompt).
- The screen may go blank for a few seconds while performing the auto-sizing function. This is not a malfunction.
- Although the signals for picture aspect ratio 5:4 (resolution: 1280 x 1024) do not fill the screen to the edges, the picture is accurately displayed.

Selecting the On-screen Display language

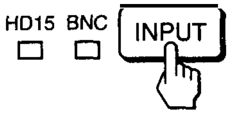
If you need to change the OSD language, see "Using the LANG (Language) On-screen Display" on page 17.
The default setting is English.

Selecting the Input Signal

This monitor has two signal input connectors (HD15 and 5 BNC) and can be connected to two computers. When the power of both computers is on, select the input signal you want to view as follows.

1 **Turn on the monitor and both computers.**

2 **Press the INPUT button to select the HD15 or 5 BNC input signal.**
Each time you press the INPUT button, the input signal and corresponding indicator alternate.



Selecting the INPUT signal mode

This monitor has two modes of input signal selection, "AUTO" and "MANUAL."

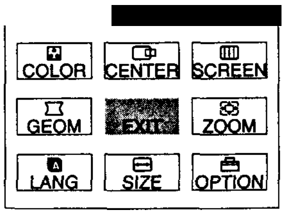
When "AUTO" is selected

If no signal is input from the selected connector, the monitor automatically selects the other connector's signal. When you restart the computer you want to view, or that computer is in power saving mode, the monitor may automatically select the other connector's signal. This is because the monitor switches from the interrupted signal to the constant signal. If this happens, manually select the desired signal using the INPUT button.

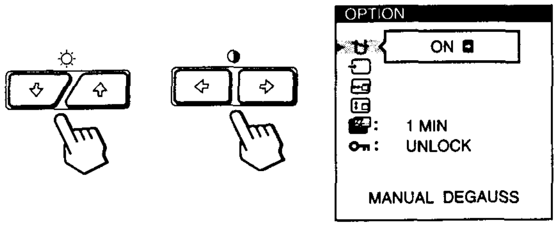
When "MANUAL" is selected

Even if no signal is input from the selected connector, the monitor does not select the other connector's signal.

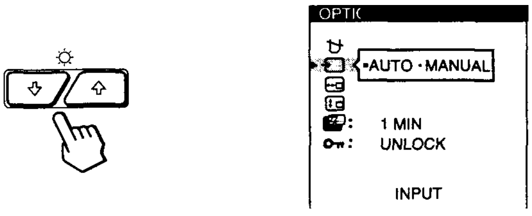
1 **Press the MENU button.**
The MENU OSD appears.



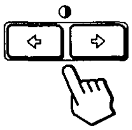
2 **Press the ☼↓/↑ and ◑←/→ buttons to select " OPTION," and-press the MENU button again.**
The OPTION OSD appears.



3 **Press the ☼↓/↑ buttons to select " (INPUT)."**



4 **Press the ◑←/→ buttons to select "AUTO" or "MANUAL."**



The OPTION OSD automatically disappears after about 30 seconds.
To close the OSD, press the MENU button again.

For more information on using the OSD, see "Introducing the On-screen Display **System"** on page 9.

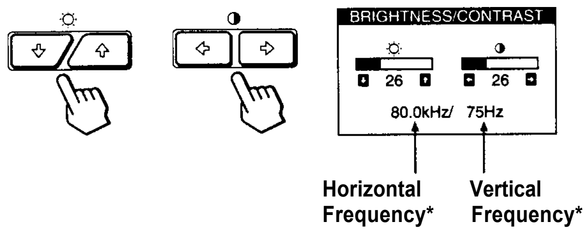
Before adjusting

- Connect the monitor and the computer, and turn them on.
- Select " LANG" in the MENU OSD, then select "ENGLISH" (see page 17).

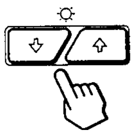
Adjusting the Picture Brightness and Contrast

Once the setting is adjusted, it will be stored in memory for all input signals received.

1 **Press the ☼ (brightness) ⬇/⬆ or ◑ (contrast) ⬅/➡ buttons.**
The BRIGHTNESS/CONTRAST OSD appears.

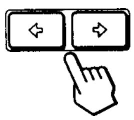


2 **For brightness adjustment**
Press the ☼⬇/⬆ buttons.



⬆ . . . for more brightness
⬇ . . . for less brightness

3 **For contrast adjustment**
Press the ◑⬅/➡ buttons.



➡ . . . for more contrast
⬅ . . . for less contrast

The OSD automatically disappears after about 3 seconds.

To reset, press the RESET button while the OSD is on. The brightness and contrast are both reset to the factory settings.

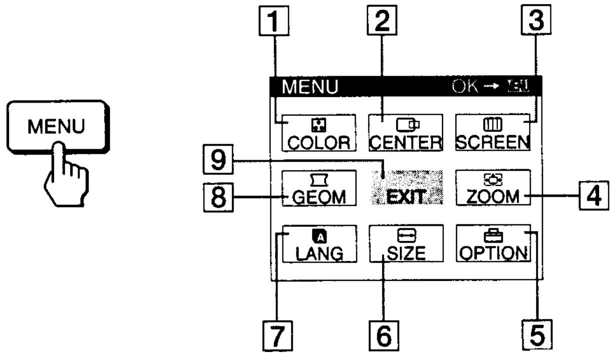
* The horizontal and vertical frequencies for the received input signal appear in the BRIGHTNESS/CONTRAST OSD.

Introducing the On-screen Display System

Most adjustments are made using the MENU OSD.

MENU OSD

Press the MENU button to display the MENU OSD.
This MENU OSD contains links to the other OSDs described below.



EN

1 **COLOR**
Displays the COLOR OSD for adjusting the color temperature.

2 **CENTER**
Displays the CENTER OSD for adjusting the centering of the picture.

3 **SCREEN**
Displays the SCREEN OSD for adjusting the vertical and horizontal convergence, etc.

4 **ZOOM**
Displays the ZOOM OSD for enlarging and reducing the picture.

5 **OPTION**
Displays the OPTION OSD for adjusting the OSD position and degaussing the screen, etc.

6 **SIZE**
Displays the SIZE OSD for adjusting the picture size.

7 **LANG**
Displays the LANGUAGE OSD for selecting the language.

8 **GEOM**
Displays the GEOMETRY OSD for adjusting the picture rotation and pincushion, etc.

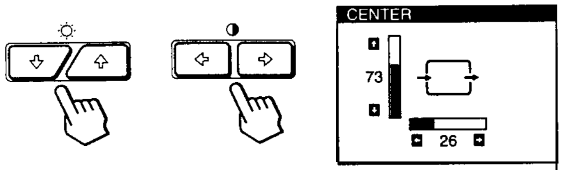
9 **EXIT**
Closes the MENU OSD.

Using the CENTER On-screen Display

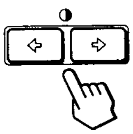
The CENTER settings allow you to adjust the centering of the picture.
Once the setting is adjusted, it will be stored in memory for the current input signal.

1 Press the MENU button.
The MENU OSD appears.

2 Press the ☼↓/↑ and ◑←/→ buttons to select " CENTER," and press the MENU button again.
The CENTER OSD appears.

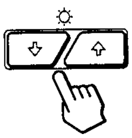


3 For horizontal adjustment
Press the ◑←/→buttons.



→ to move the picture right
← to move the picture left

4 For vertical adjustment
Press the ☼↓/↑ buttons.



↑ . . . to move the picture up
↓ . . . to move the picture down

The OSD automatically disappears after about 30 seconds.
To close the OSD, press the MENU button again.

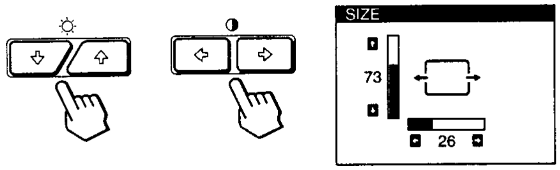
To reset, press the RESET button while the OSD is on.
The horizontal and vertical centerings are both reset to the factory settings.

Using the SIZE On-screen Display

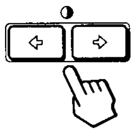
The SIZE settings allow you to adjust the size of the picture. Once the setting is adjusted, it will be stored in memory for the current input signal.

1 Press the MENU button.
The MENU OSD appears.

2 Press the ☼↓/↑ and ◑←/→ buttons to select " SIZE," and press the MENU button again.
The SIZE OSD appears.

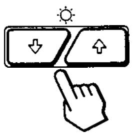


3 For horizontal adjustment
Press the ◑←/→ buttons.



→ to increase picture size
← to decrease picture size

4 For vertical adjustment
Press the ☼↓/↑ buttons.



↑ . . . to increase picture size
↓ . . . to decrease picture size

The OSD automatically disappears after about 30 seconds.
To close the OSD, press the MENU button again.

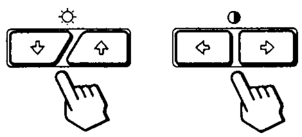
To reset, press the RESET button while the OSD is on.
The horizontal and vertical sizes are both reset to the factory settings.

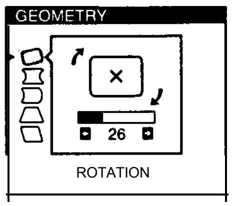
Using the GEOM (Geometry) On-screen Display

The GEOM (geometry) settings allow you to adjust the shape and orientation of the picture.
Once the rotation is adjusted, it will be stored in memory for all input signals received. All other adjustments will be stored in memory for the current input signal.

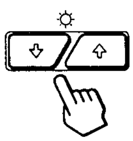
1 Press the MENU button.
The MENU OSD appears.

2 Press the ☼↓/↑ and ◑←/→ buttons to select " GEOM," and press the MENU button again.
The GEOMETRY OSD appears.



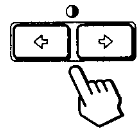


3 Press the ☼↓/↑ buttons to select the item you want to adjust.



Select	To
ROTATION	adjust the picture rotation
PINCUSHION	adjust the picture sides
PINBALANCE	adjust the picture side balance
KEYSTONE	adjust the picture width
KEY BALANCE	adjust the picture shape balance

4 Press the ◑←/→ buttons to adjust the settings.



For	Press
ROTATION	→ . . . to rotate the picture clockwise ← . . to rotate the picture counterclockwise
PINCUSHION	→ . . . to expand the picture sides ← . . to contract the picture sides
PINBALANCE	→ . . . to move the picture sides to the right ← . . . to move the picture sides to the left
KEYSTONE	→ . . . to increase the picture width at the top ← . . . to decrease the picture width at the top
KEYBALANCE	→ to move the top of the picture to the right ← . . . to move the top of the picture to the left

The OSD automatically disappears after about 30 seconds.
To close the OSD, press the MENU button again.

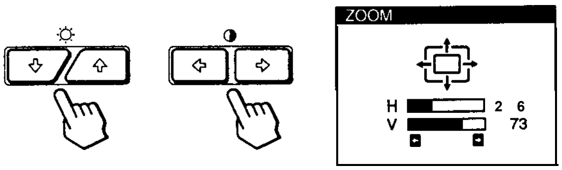
To reset, press the RESET button while the OSD is on.
The selected item is reset to the factory setting.

Using the ZOOM On-screen Display

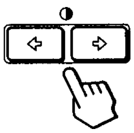
The ZOOM settings allow you to enlarge or reduce the picture.
Once the setting is adjusted, it will be stored in memory for the current input signal.

1 Press the MENU button.
The MENU OSD appears.

2 Press the ☼↓/↑ and ◑←/→ buttons to select "ZOOM," and press the MENU button again.
The ZOOM OSD appears.



3 Press the ◑←/→buttons to adjust the picture zoom.



→ . . . to enlarge the picture
← . . . to reduce the picture

The OSD automatically disappears after about 30 seconds. To close the OSD, press the MENU button again.

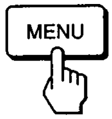
To reset, press the RESET button while the OSD is on.

Note
The picture zoom adjustment will stop as soon as either the horizontal or vertical size reaches its maximum or minimum value.

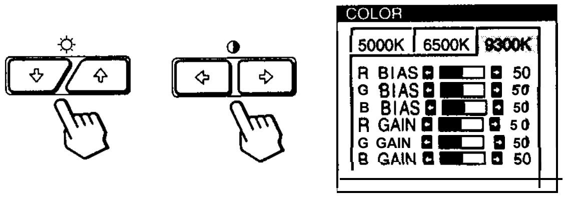
Using the COLOR On-screen Display

You can change the monitor' s color temperature. For example, you can adjust or change the colors of a picture on the screen to match the actual colors of the printed picture. Once the setting is adjusted, it will be stored in memory for all input signals received.

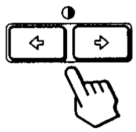
1 Press the MENU button.
The MENU OSD appears.



2 Press the ☼↓/↑ and ◑←/→ buttons to select " COLOR," and press the MENU button again.
The COLOR OSD appears.



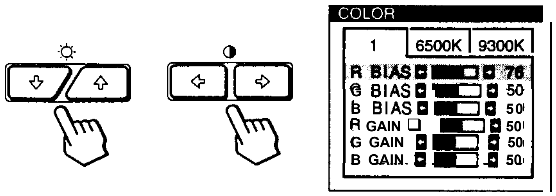
3 Press the ◑←/→ buttons to select the color temperature.



There are three color temperature modes in the OSD.
The preset adjustments are:
5000K, 6500K, 9300K

4 Fine tuning the color temperature

Press the ☼⬇/⬆ buttons to select an item and adjust by pressing the ◑⬅/➡ buttons.



Select **R (red), G (green), or B (blue) BIAS** to adjust the black level of each color' s signal.

Select **R (red), G (green), or B (blue) GAIN** to adjust the white level of each color' s signal.

The "5000K," "6500K" or "9300K" disappears and the new color settings are memorized for each of the three color modes.
The color temperature modes change as follows:
5000K+, 1, 6500K+, 2, 9300K+, 3

The OSD automatically disappears after about 30 seconds. To close the OSD, press the MENU button again.

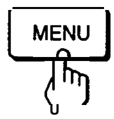
To reset, press the RESET button while the OSD is on. The selected item is reset to the factory settings.

Using the SCREEN On-screen Display

You can adjust convergence settings to eliminate red or blue shadows that may appear around objects on the screen.
Adjust the CANCEL MOIRE function to eliminate wavy or elliptical patterns that may appear on the screen.
Adjust the LANDING function to correct color imbalances at the four comers of the screen due to influence from the earth' s magnetism.
Once CANCEL MOIRE is adjusted, it will be stored in memory for the current input signal. All other adjustments will be stored in memory for all input signals received.

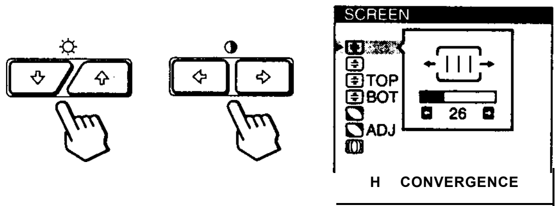
1 Press the MENU button.

The MENU OSD appears.



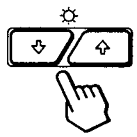
2 Press the ☼⬇/⬆ and ◑⬅/➡ buttons to select " SCREEN," and press the MENU button again.

The SCREEN OSD appears.



EN

3 Press the ☼⬇/⬆ buttons to select the item you want to adjust.



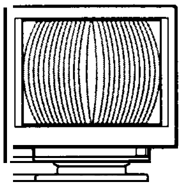
Select	To
H CONVERGENCE	adjust the horizontal convergence
V CONVERGENCE	adjust the vertical convergence
TOP V CONVER TOP	adjust the screen' s upper vertical convergence
BOT V CONVER BOTTOM	adjust the screen's lower vertical convergence

(continued)

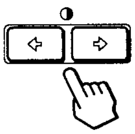
Select	To
LANDING	select one of the four comers that needs color correction due to influence from the earth' s magnetism
ADJ LANDING ADJUST	correct the color at one of the four comers of the screen
CANCEL MOIRE *	turn the moire cancellation function "ON" or "OFF." CANCEL MOIRE must be "ON" for " ADJ (MOIRE ADJUST)" to appear on the screen.
ADJ MOIRE ADJUST	adjust the degree of moire cancellation

* Moire is a type of natural interference which produces soft or wavy lines on your screen. It may appear due to interference between the regulated pattern of the picture from the input signal and the phosphor pitch pattern of the CRT.

Example of moire:



4 Press the ◑←/→ buttons to adjust the settings.



For	Press
H CONVERGENCE	→ . . . to shift red shadows to the right and blue shadows to the left ← . . to shift red shadows to the left and blue shadows to the right
V CONVERGENCE	→ . . . to shift red shadows up and blue shadows down ← . . . to shift red shadows down and blue shadows up
TOP V CONVER TOP	→ . . to shift red shadows up and blue shadows down ← . . to shift red shadows down and blue shadows up
BOT V CONVER BOTTOM	→ . . . to shift red shadows up and blue shadows down ← . . to shift red shadows down and blue shadows up
LANDING	→ or ← . . . to select the comer of the screen you want to adjust : top left : top right : bottom left : bottom right
ADJ LANDING ADJUST	→ or ← to reduce any irregularities in the color to a minimum
CANCEL MOIRE	→ . to turn CANCEL MOIRE "ON" ← . to turn CANCEL MOIRE "OFF"
ADJ MOIRE ADJUST	→ or ← . to adjust the screen until the moire is at a minimum

Note

The picture may become fuzzy when CANCEL MOIRE is set to "ON."

The OSD automatically disappears after about 30 seconds. To close the OSD, press the MENU button again.

To reset, press the RESET button while the OSD is on. The selected item is reset to the factory setting.

Using the OPTION On-screen Display

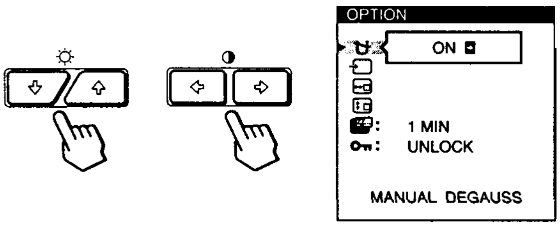
The OPTION OSD allows you to manually degauss the screen and adjust settings such as the OSD position and power saving delay time. It also allows you to lock the controls.

Degaussing the screen

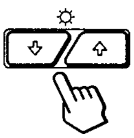
The monitor screen is automatically degaussed (demagnetized) when the power is turned on.
You can also manually degauss the monitor.

1 **Press the MENU button.**
The MENU OSD appears.

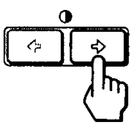
2 **Press the ☼↓/↑ and ◑←/→ buttons to select " OPTION," and press the MENU button again.**
The OPTION OSD appears.



3 **Press the ☼↓/↑ buttons to select " (MANUAL DEGAUSS)."**



4 **Press the ◑→ button.**
The screen is degaussed for about 3 seconds.



If you need to degauss the screen a second time, wait for at least 20 minutes before repeating the steps above.

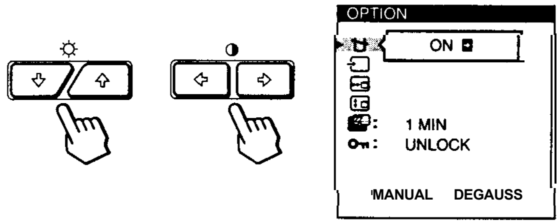
The OPTION OSD automatically disappears after about 30 seconds.
To close the OSD, press the MENU button again.

Changing the on-screen display position

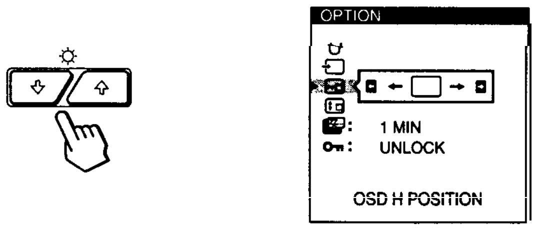
You can change the OSD position (for example, when you want to adjust the picture behind the OSD).

1 **Press the MENU button.**
The MENU OSD appears.

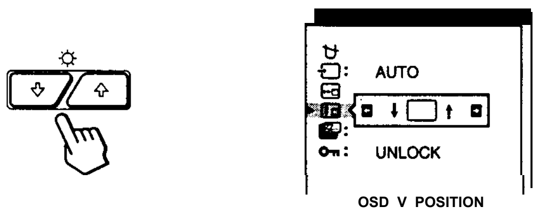
2 **Press the ☼↓/↑ and ◑←/→ buttons to select " OPTION," and press the MENU button again.**
The OPTION OSD appears.



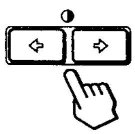
3 **Press the ☼↓/↑ buttons to select " (OSD H POSITION)" or " (OSD V POSITION)."**
Select " (OSD H POSITION)" to adjust the horizontal position.



Select " (OSD V POSITION)" to' adjust the vertical position.



4 **Press the ◑←/→ buttons to move the OSD to the desired position.**



The OPTION OSD automatically disappears after about 30 seconds.
To close the OSD, press the MENU button again.

To reset, press the RESET button while the OSD is on.

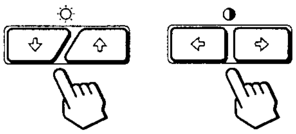
EN

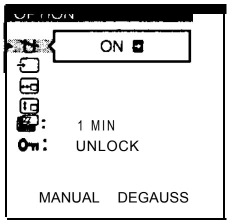
Setting the power saving delay time

You can set the delay time before the monitor enters the power saving mode. See page 19 for more information on this monitor's power saving capabilities.

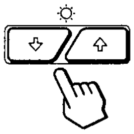
1 **Press the MENU button.**
The MENU OSD appears.

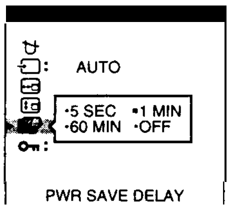
2 **Press the ☼↓/↑ and ◑←/→ buttons to select " OPTION," and press the MENU button again.**
The OPTION OSD appears.



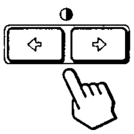


3 **Press the ☼↓/↑ buttons to select " (PWR SAVE DELAY)."**





4 **Press the ◑←/→ buttons to select the desired time.**



When PWR SAVE DELAY is set to "OFF," the monitor does not go into power saving mode.

The OPTION OSD automatically disappears after about 30 seconds.
To close the OSD, press the MENU button again.

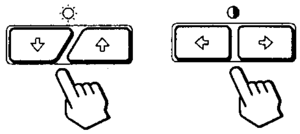
To reset, press the RESET button while the OSD is on.

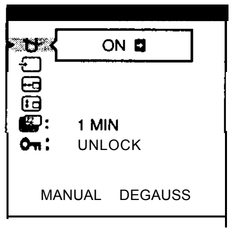
Locking the controls

The control lock function disables all of the buttons on the front panel except the ⏻ (power) switch and MENU button.

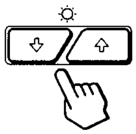
1 **Press the MENU button.**
The MENU OSD appears.

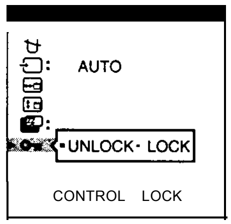
2 **Press the ☼↓/↑ and ◑←/→ buttons to select " OPTION," and press the MENU button again.**
The OPTION OSD appears.



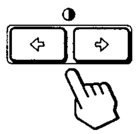


3 **Press the ☼↓/↑ buttons to select " (CONTROL LOCK)."**





4 **Press the ◑←/→ buttons to select "LOCK."**



The OPTION OSD automatically disappears after about 30 seconds.
To close the OSD, press the MENU button again.

Once you select "LOCK," you cannot select any items except "EXIT" and " OPTION" in the MENU OSD.
If you press any button other than the ⏻ (power) switch and MENU button, the mark appears on the screen.

To cancel the control lock

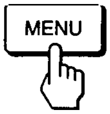
Repeat steps 1 through 3 above and press the ◑←/→ buttons to select "UNLOCK."

Using the LANG (language) On-screen Display

English, French, German, Spanish, Italian and Japanese versions of the OSDs are available.

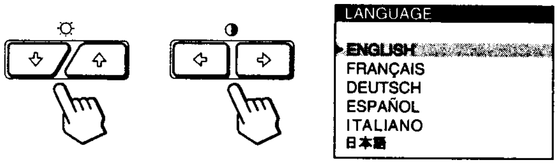
1 Press the MENU button.

The MENU OSD appears.

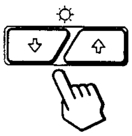


2 Press the ☼↓/↑ and ◑←/→ buttons to select "🅰 LANG," and press the MENU button again.

The LANGUAGE OSD appears.



3 Press the ☼↓/↑ buttons to select the desired language.



ENGLISH: English, FRANÇAIS: French,
DEUTSCH: German, ESPAÑOL: Spanish,
ITALIANO: Italian, or 日本語 : Japanese.

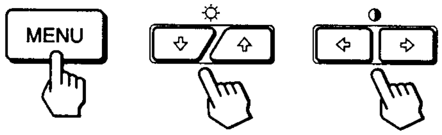
The OSD automatically disappears after about 30 seconds. To close the OSD, press the MENU button again.

To reset to English, press the RESET button while the OSD is on.

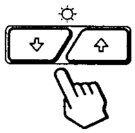
Resetting the Adjustments

Resetting an adjustment item

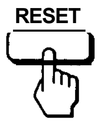
1 Press the MENU, ☼↓/↑ and ◑←/→ buttons to select the OSD containing the item you want to reset.



2 Press the ☼↓/↑ buttons to select the item you want to reset.



3 Press the RESET button.

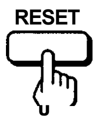


EN

Resetting all of the adjustment data for the current input signal

When there is no OSD displayed, press the RESET button.

All of the adjustments data for the current input signal is reset to the factory settings.
Note that adjustment data not affected by changes in input signal (OSD language, OSD position, input signal selection, power saving delay time and the control lock function) is not reset to the factory settings.



Resetting all of the adjustment data for all input signals

Press and hold the RESET button for more than two seconds.

All of the adjustment data, including the brightness and contrast, is reset to the factory settings.

Preset and User Modes

This monitor has factory preset modes for the most popular industry standards for true “plug and play” compatibility.

When a new input signal is entered, the monitor selects the appropriate factory preset mode and momentarily adjusts the phase calibration to provide a high quality picture to the center of the screen. The calibration is stored in memory and is immediately recalled whenever the same input signal is received.

No.	Resolution (dots x lines)	Horizontal Frequency	Vertical Frequency	Graphics Mode
1	640 x 350	31.5 kHz	70 Hz	MCGA
2	640 x 480	31.5 kHz	60 Hz	VGA-G
3	640 x 480	37.5 kHz	75 Hz	EVGA
4	640 x 480	43.3 kHz	85 Hz	VESA
5	720 x 400	31.5 kHz	70 Hz	VGA-Text
6	720 x 400	37.9 kHz	85 Hz	VESA
7	800 x 600	37.9 kHz	60 Hz	SVGA
8	800 x 600	46.9 kHz	75 Hz	ESVGA
9	800 x 600	53.7 kHz	85 Hz	VESA
10	832 x 624	49.7 kHz	75 Hz	Macintosh 16″ Color
11	1024 x 768	48.4 kHz	60 Hz	VESA
12	1024 x 768	56.5 kHz	70 Hz	VESA
13	1024 x 768	60.0 kHz	75 Hz	EUVGA
14	1024 x 768	60.2 kHz	75 Hz	Macintosh 19″ Color
15	1024 x 768	68.7 kHz	85 Hz	VESA
16	1152 x 864	67.5 kHz	75 Hz	VESA
17	1152 x 870	68.7 kHz	75 Hz	Macintosh 21″ Color
18	1280 x 960	60.0 kHz	60 Hz	VESA
19	1280 x 960	85.9 kHz	85 Hz	VESA
20	1280 x 1024	64.0 kHz	60Hz	VESA
21	1280 x 1024	80.0 kHz	75 Hz	VESA
22	1280 x 1024	91.1 kHz	85 Hz	VESA
23	1600 x 1200	75.0 kHz	60Hz	VESA
24	1600x1200	81.3kHz	65Hz	VESA
25	1600 x 1200	87.5 kHz	70 Hz	VESA
26	1600 x 1200	93.8 kHz	75 Hz	VESA
27*	1600 x 1200	106.3 kHz	85 Hz	VESA

* GDM500PS only

For input signals that do not match one of the factory preset modes, the digital Multiscan technology of this monitor performs all of the adjustments necessary to ensure that a clear picture appears on the screen for any timing in the monitor’ s frequency range. However, it may be necessary to fine tune the vertical/horizontal size and centering. Simply press the ASC button or adjust the monitor according to the adjustment instructions. The adjustments are stored automatically as a user mode and recalled whenever the corresponding input signal is received.

Recommended horizontal and vertical timing conditions

Horizontal sync width duty should be: >4.8% of total horizontal time.
Horizontal blanking width should be: >2.5 μsec.
Vertical blanking width should be: > 450 μsec.

Note for Windows users

For Windows users, check your video board manual or the utility program which comes with your graphic board and select the highest available refresh rate to maximize monitor performance.

Adjusting the monitor’ s resolution and color number

Adjust the monitor’ s resolution and color number by referring to your computer’ s instruction manual. The color number may vary according to your computer or video board. The color palette setting and the actual number of colors are as follows:

- High Color (16 bit) → 65,536 colors
- True Color (24 bit) → about 16.77 million colors

In true color mode (24 bit), speed may be slower.

Power Saving Function

This monitor meets the power-saving guidelines set by VESA and Energy Star, as well as the more stringent NUTEK .
If the monitor is connected to a computer or video graphics board that is VESA DPMS (Display Power Management Signaling) compliant, the monitor will automatically reduce power consumption in three stages as shown below.

You can set the delay time before the monitor enters the power saving mode using the OSD. Set the time according to "Setting the power saving delay time" on page 16.

Note

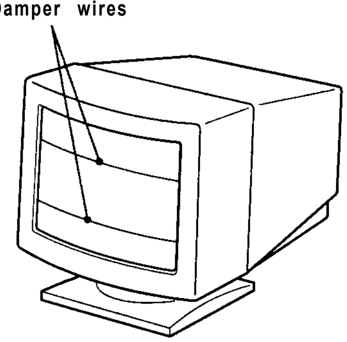
If no video signal is input to the monitor, the "NO INPUT SIGNAL" message (page 20) appears. After the delay time has passed, the power saving function automatically puts the monitor into the active-off mode and the ⏻ indicator lights up orange. Once the horizontal and vertical sync signals are detected, the monitor automatically resumes its normal operation mode.

	Power consumption mode	Screen	Horizontal sync signal	Vertical sync signal	Power consumption	Recovery time	⏻ indicator
1	Normal operation	active	present	present	≤ 145 W (GDM-500PS) ≤ 125 W (GDM-400PS)	—	Green
2	Standby (1st mode)	blank	absent	present	Approx. 72 W (GDM-500PS) Approx. 65 W (GDM-400PS)	Approx. 3 sec.	Green and orange alternate
3	Suspend (2nd mode)	blank	present	absent	Approx. 8 W	Approx. 3 sec.	Green and orange alternate
4	Active-off (3rd mode)	blank	absent	absent	Approx. 4 W	Approx. 10 sec.	Orange
5	Power-off	—	—	—	0 w	—	Off

EN

Damper Wires

When viewing a white background, very thin horizontal lines may be visible on the screen as shown below. These lines are the shadows of the damper wires and are characteristic of CRTs that use aperture grilles. The wires are attached to the aperture grille on the inside of the Trinitron tube and prevent the vibration of the aperture grille.



Plug & Play

This monitor complies with the DDC™1, DDC2B, DDC2AB and DDC2B+ Display Data Channel (DDC) standards of VESA.
When a DDC1 host system is connected, the monitor synchronizes with the V. CLK in accordance with the VESA standards and outputs the EDID (Extended Display Identification Data) to the data line.
When a DDC2B, DDC2AB or DDC2B+ host system is connected, the monitor automatically switches to the appropriate standard.

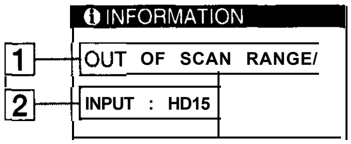
DDC™ is a trademark of the Video Electronics Standard Association.

Note

When using Windows 95/98, the DDC standard does not apply to the 5 BNC connectors. If you use the DDC standard, connect the HD15 connector to the computer with the supplied video signal cable.

Warning Messages

If there is something wrong with the input signal, one of the following messages appears.



1 The input signal condition

"OUT OF SCAN RANGE" indicates that the input signal is not supported by the monitor's specifications.

"NO INPUT SIGNAL" indicates that no signal is input, or the input signal from the selected input connector is not received.

2 The selected input connector

Indicates which input connector is receiving the wrong signal. If there is something wrong with the signal from both input connectors, "HD15" and "BNC" are displayed alternately.

To solve these problems, see "Troubleshooting" below.

Troubleshooting

This section may help you isolate the cause of a problem and as a result, eliminate the need to contact technical support.

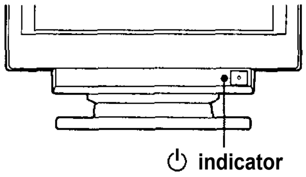
Symptom	Check these items
No picture	
If the ⏻ indicator is not lit	• Check that the power cord is properly connected. • Check that the ⏻ (power) switch is in the "on" position.
If the "NO INPUT SIGNAL" message appears on the screen, or if the ⏻ indicator is either orange or alternating between green and orange	• The screen is blank when the monitor is in power saving mode. Try pressing any key on the computer keyboard. • Check that your computer power switch is in the "on" position. • Check that the input select **setting** is correct. • Check that the video signal cable is properly connected and all plugs are firmly seated in their sockets. • Check that the 5 BNCs are connected in the correct order (from left to right: Red-Green-Blue-HD-VD) (page 6). • Ensure that no pins are bent or pushed in the HD15 video input connector. • Check that the video board is completely seated in the proper bus slot.
If the "OUT OF SCAN RANGE" message appears on the screen	• Check that the video frequency range is within that specified for the monitor. Horizontal: 30 – 94 kI-Iz (GDM-400PS), 30 – 107 kHz (GDM-500PS) Vertical: 48 – 160 Hz Refer to your computer's instruction manual to adjust the video frequency range. • If you are using a video signal cable adapter, check that it is correct.
If no message is displayed and the ⏻ indicator is green or flashing orange	• See "Self-diagnosis Function" (page 22).
If using a Macintosh system	• Check that the Macintosh adapter and the video signal cable are properly connected (page 6).
If using Windows 95 / 98	• If you replaced an old monitor with this monitor, reconnect the old monitor and do the following. Install the Windows Monitor Information Disk (page 7) and select "GDM-400PS" or "GDM-500PS" from among the Sony monitors in the Windows 95/ 98 monitor selection screen. If you choose to select "Plug and Play," connect the monitor to the computer with the HD15 video signal. You cannot use the five BNC connectors.
Picture is scrambled	• Check your graphics board manual for the proper monitor setting. • Check this manual and confirm that the graphics mode and the frequency you are trying to operate at is supported. Even if the frequency is within the proper range, some video boards may have a sync pulse that is too narrow for the monitor to sync correctly.

Symptom	Check these items
Color is not uniform	• Degauss the monitor (page 15). If you place equipment which generates a magnetic field, such as a loudspeaker, near the monitor, or you change the direction of the monitor, color may lose uniformity. The degauss function demagnetizes the metal frame of the CRT to obtain a neutral field for uniform color reproduction. If a second degauss cycle is needed, allow a minimum interval of 20 minutes for the best result. • Adjust the landing (pages 13 - 14).
You cannot adjust the monitor with the buttons on the front panel	• If the control lock function is set to on, set it to off using the OPTION OSD (page 16).
White does not look white	• Adjust the color temperature (pages 12 – 13). • Check that the 5 BNCs are connected in the correct order (from left to right: RedGreen-Blue-HD-VD) (page 6).
Screen image is not centered or sized properly	• Press the ASC button (page 7). • Adjust the size or centering (page 10). • Some video modes do not fill the screen to the edges. This problem tends to occur with certain video boards.
Edges of the image are curved	• Adjust the geometry (page 11).
White lines show red or blue shadows at edges	• Adjust the convergence (pages 13 – 14).
Picture is fuzzy	• Adjust the contrast and brightness (page 9). • Degauss the monitor (page 15). If you place equipment which generates a magnetic field, such as a loudspeaker, near the monitor, or you change the direction of the monitor, color may lose uniformity. The degauss function demagnetizes the metal frame of the CRT to obtain a neutral field for uniform color reproduction. If a second degauss cycle is needed, allow a minimum interval of 20 minutes for the best result. • If red or blue shadows appear along the edges of images, adjust the convergence (pages 13 – 14). • If the moire is cancelled, the picture may become fuzzy. Decrease the moire cancellation effect (pages 13 – 14).
Picture bounces or has wavy oscillations	• Isolate and eliminate any potential sources of electric or magnetic fields. Common causes for this symptom are electric fans, fluorescent lighting or laser printers. • If you have another monitor close to this monitor, increase the distance between them to reduce the interference. • Try plugging the monitor into a different AC outlet, preferably on a different circuit; • Try the monitor on a different computer in a different room.
Picture is flickering	• Set the refresh rate on the computer to obtain the best possible picture by referring to the computer' s manual.
Picture appears to be ghosting	• Eliminate the use of video cable extensions and/or video switch boxes if this symptom occurs. Excessive cable length or a weak connection can produce this symptom.
Wavy or elliptical (moire) pattern is visible	• Cancel the moire (pages 13 – 14). The moire may be modified depending on the connected computer. • Due to the relationship between resolution, monitor dot pitch and the pitch of some image patterns, certain screen backgrounds sometimes show moire. Change your desktop pattern.
Two fine horizontal lines (wires) are visible	• These wires stabilize the vertically striped aperture grille (page 19). This aperture grille allows more light to pass through to the screen giving the Trinitron CRT more color and brightness.
Hum is heard right after the power is turned on	• When the power is turned on, the auto-degauss cycle is activated. While the auto-degauss cycle is activated (3 seconds), a hum may be heard. The same hum is heard when the monitor is manually degaussed. This is not a malfunction.

EN

Self-diagnosis Function

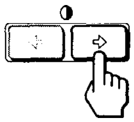
This monitor is equipped with a self-diagnosis function. If there is a problem with your monitor or computer(s), the screen will go blank and the ⏻ indicator will either light up green or flash orange.



If the ⏻ indicator is green

1 Remove any plugs from the video input 1 and 2 connectors, or turn off the connected computer(s).

2 Press and hold the ◑➡ button for 2 seconds.



If all four color bars appear (white, red, green, blue), the monitor is working properly. Reconnect the video input cables and check the condition of your computer(s).

If the color bars do not appear, there is a potential monitor failure. Inform your authorized Sony dealer of the monitor' s condition.

If the ⏻ indicator is flashing orange

Press the ⏻ button to turn the monitor off and on.

If the ⏻ indicator lights up green, the monitor is working properly.

If the ⏻ indicator is still flashing, there is a potential monitor failure. Count the number of seconds between orange flashes of the ⏻ indicator and inform your authorized Sony dealer of the monitor' s condition. Be sure to note the model name and serial number of your monitor. Also note the make and model of your computer and video board.

Specifications

GDM4OOPS

Picture tube	0.25 – 0.27 mm aperture grille pitch 19 inches measured diagonally 90-degree deflection
Viewable image size	Approx. 365 x 273 mm (w/h) (14 3/8 x10 3/4inches) 18.0" viewing image
Resolution	Horizontal: Max. 1600 dots Vertical: Max. 1200 lines
Standard image area	Approx. 330 x 264 mm (w/h) (13 x 10 1/2 inches) or Approx. 352 x 264 mm (w/h) (13 7/8 X 10 1/2 inches)
Deflection frequency	Horizontal: 30 to 94 kHz Vertical: 48 to 160 Hz
AC input voltage/current	100 to 240 V, 50 – 60 Hz, 1.8 – 1.0 A
Power consumption	Max. 125 W
Dimensions	444x467x453mm(w/h/d) (17 1/2 x 18 1/2 x 17 7/8 inches)
Mass	Approx. 25 kg (55 lb 2 oz)
Supplied accessories	See page 6

GDM-500PS

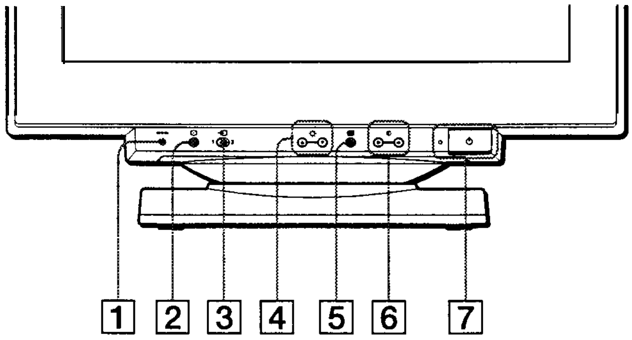
Picture tube	0.25 – 0.27 mm aperture grille pitch 21 inches measured diagonally 90-degree deflection
Viewable image size	Approx. 403.8 x 302.2 mm (w/h) (16 x 12 inches) 19.8" viewing image
Resolution	Horizontal: Max. 1600 dots Vertical: Max. 1200 lines
Standard image area	Approx. 388 x 291 mm (w/h) (15 3/8 x 11 1/2 inches) or Approx. 364 x 291 mm (w/h) (14 3/8 X 11 1/2 inches)
Deflection frequency	Horizontal: 30 to 107 kHz Vertical: 48 to 160 Hz
AC input voltage/ current	100 to 240 V, 50 – 60 Hz, 2.0 – 1.0 A
Power consumption	Max. 145 W
Dimensions	498x505x474mm(w/h/d) (19 5/8 x 20 x 18 3/4 inches)
Mass	Approx. 31 kg (68 lb 5 oz)
Supplied accessories	See page 6

Design and specifications are subject to change without notice.

Identifying parts and controls

See the pages in parentheses for further details.

Front



1 **→•←-(reset) button (page 14)**
This button resets the adjustments to the factory settings.

2 **ASC (auto sizing and centering) button (page 7)**
This button automatically adjusts the size and centering of the picture.

3 **(input) switch (page 7)**
This switch selects the INPUT 1 (video input 1 connector: 1) or INPUT 2 (video input 2 connector: 2) video input signal.

4 **☼ (brightness) ↓/↑ buttons (page 9)**
These buttons display the BRIGHTNESS/CONTRAST menu and function as the ↓/↑ buttons when selecting menu items.

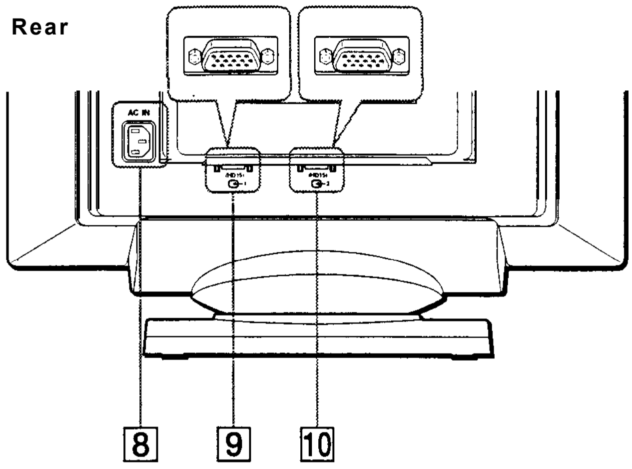
5 **(menu) button (page 9)**
This button displays the main menu.

6 **◐ (contrast) ←/→ buttons (page 9)**
These buttons display the BRIGHTNESS/CONTRAST menu and function as the ←/→ buttons when making adjustments.

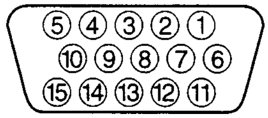
7 **⏻ (power) switch and indicator (pages 6, 14, 18)**
This button turns the monitor on and off. The power indicator lights up in green when the monitor is turned on, and either flashes in green and orange, or lights up in orange when the monitor is in power saving mode.

8 **AC IN connector (page 6)**
This connector provides AC power to the monitor.

Rear



9 **Video input 1 connector (HD15) (1) (page 6)**
This connector inputs RGB video signals (0.700 Vp-p, positive) and sync signals.



Pin No.	Signal
1	Red
2	Green (Sync on Green)
3	Blue
4	ID (Ground)
5	DDC Ground*
6	Red Ground
7	Green Ground
8	Blue Ground
9	DDC + 5V*
10	Ground
11	ID (Ground)
12	Bi-Directional Data (SDA)*
13	H. Sync
14	V. Sync
15	Data Clock (SCL)*

* DDC (Display Data Channel) is a standard of VESA.

10 **Video input 2 connector (HD15) (2) (page 6)**
This connector inputs RGB video signals (0.700 Vp-p, positive) and sync signals. Pin assignment is the same as **9** described above.

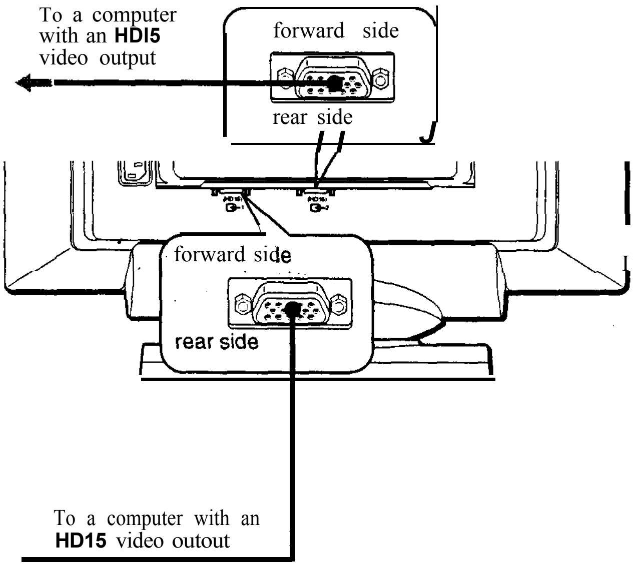
GB

Setup

This monitor works with platforms running at horizontal frequencies between 30 and 121 kHz.

Step 1: Connect your monitor to your computer

With the monitor and computer switched off. connect the video signal cable to the video input connector of the monitor and to the video output of the computer.

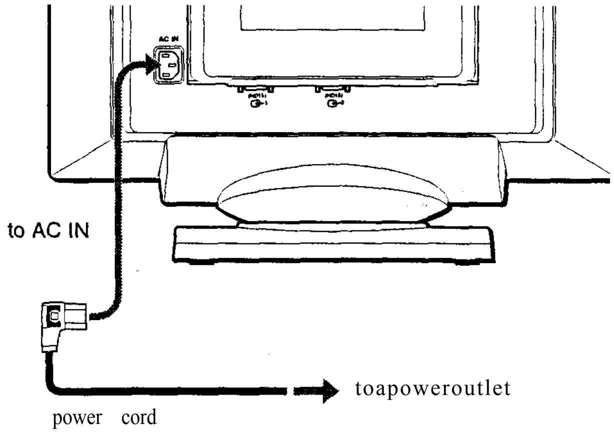


Notes

- Do not touch the pins of the video signal cable connecto r as this might bend the pins.
- When connecting the video signal cable, check the alignment of the HDIS connector. Do not force the connector in the wrong way or the pins might bend.

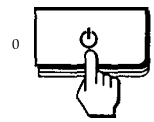
Step 2: Connect the power cord

With the monitor and computer switched off, first connect the proper power cord for your local power supply to the monitor, then connect it to a power outlet.



Step 3:Turn on the monitor and computer

First turn on the monitor, then turn on the computer.



The installation of your monitor is complete.
If necessary, use the monitor's controls to adjust the picture.

If no picture appears on your screen

- Check that the monitor is correctly connected to the computer.
- If NO INPUT SIGNAL appears on the screen, follow the on-screen messages (page 1.5).
- If you are replacing an old monitor with this model and OUT OF SCAN RANGE appears on the screen, reconnect the old monitor. Then adjust the computer's graphic board so that the horizontal frequency is between 30 - 121 kHz, and the vertical frequency is between 48 - 160 Hz.

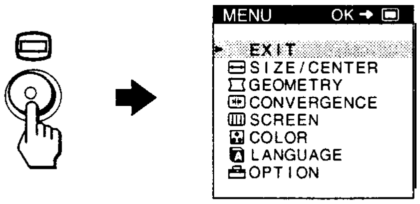
For more information about the on-screen messages, see "Trouble symptoms and remedies" on page 16.

Selecting the on-screen menu language (LANGUAGE)

English. French. German. Spanish. Italian, Dutch, Swedish, Russian and Japanese versions of the on-screen menus are available. The default setting is English.

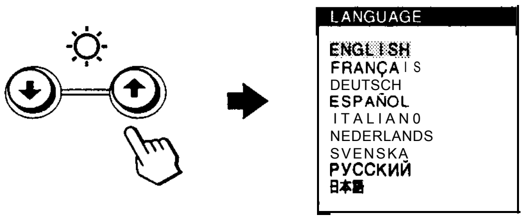
1 Press the button.

See page 9 for more information on using the button,



2 Press the ↓/↑ buttons to highlight LANGUAGE and press the button again.

See page 9 for more information on using the ↓/↑ buttons.



3 Press the ↓/↑ buttons to select a language.

- ENGLISH
- FRANCAIS: French
- DEUTSCH: German
- ESPAÑOL: Spanish
- ITALIANO: Italian
- NEDERLANDS: Dutch
- SVENSKA: Swedish
- РУССКИЙ : Russian
- 日本語: Japanese

To close the menu

Press the button once to return to the main MENU, and twice to return to normal viewing. If no buttons are pressed, the menu closes automatically after about 30 seconds.

To reset to English

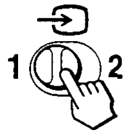
Press the (reset) button while the LANGUAGE menu is displayed on the screen.

Selecting the input signal

You can connect two computers to this monitor using the video input 1 (1) and video input 2 (2) connectors. To select one of the two computers, use the switch.

Move the switch.

The selected connector appears on the screen for 3 seconds.



"INPUT 1" (video input 1 connector: 1) or "INPUT 2" (video input 2 connector: 2) appears on the screen.

Note

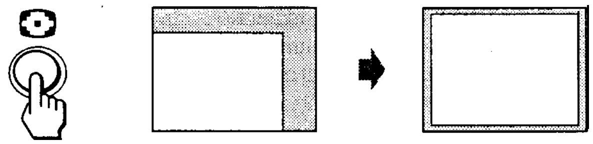
If no signal is input to the selected connector, NO INPUT SIGNAL appears on the screen. After a few seconds, the monitor enters the power saving mode. If this happens, switch to the other connector.

Automatically sizing and centering the picture

You can easily adjust the picture to fill the screen by pressing the (auto sizing and centering) button.

Press the button.

The picture automatically fills the screen,



Notes

- This function is intended for use with a computer that provides a full screen picture. It may not work properly if the background color is dark or if the input picture does not fill the screen to the edges.
- Pictures with an aspect ratio of 5:4 (resolution: 1280 x 1024, 1600 x 1280) are displayed at their actual resolution and do not fill the screen to the edges.
- The displayed image moves for a few seconds when the button is pressed. This is not a malfunction.

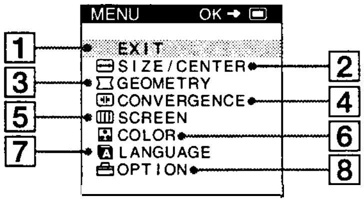
GB

Customizing Your Monitor

You can make numerous adjustments to your monitor using the on-screen menu.

Navigating the menu

Press the ⊜ button to display the main MENU on your screen.



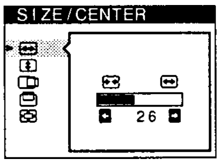
Use the ↓/↑ and ⊜ buttons to select one of the following menus. See page 9 for more information on using the ↓/↑ and ⊜ buttons.

1 EXIT

Select EXIT to close the menu.

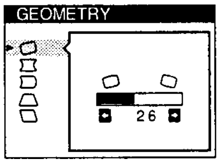
2 SIZE/CENTER (page 10)

Selects the SIZE/CENTER menu to adjust the picture's size, centering or zoom.



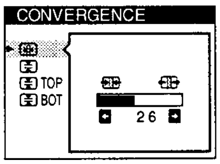
3 GEOMETRY (page 10)

Select the GEOMETRY menu to adjust the picture's rotation and shape.



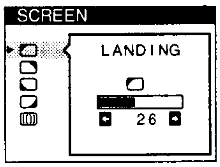
4 CONVERGENCE (page 11)

Select the CONVERGENCE menu to adjust the picture's horizontal and vertical convergence.



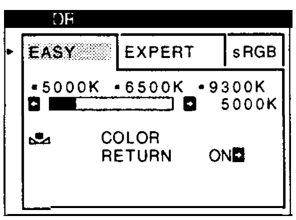
5 SCREEN (page 11)

Select the SCREEN menu to adjust the picture's quality. You can adjust the landing and moire cancellation effect.



6 COLOR (page 12)

Select the COLOR menu to adjust the picture's color temperature. You can use this to match the monitor's colors to a printed picture's colors.



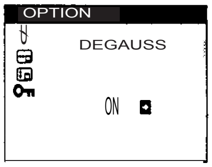
7 LANGUAGE (page 7)

Select LANGUAGE to choose the on-screen menu's language.

8 OPTION (page 13)

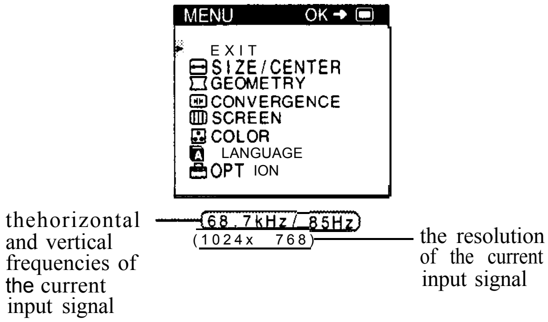
Select OPTION to adjust the monitor's options. The options include:

- degaussing the screen
- changing the on-screen menu position
- locking the controls



■ Displaying the current input signal

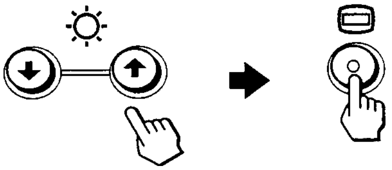
The horizontal and vertical frequencies of the current input signal are displayed in the main MENU. If the signal matches one of this monitor's factory preset modes. the resolution is also displayed.



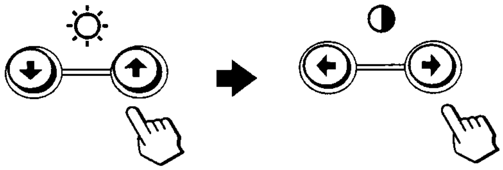
● Using the ⊟, ↓/↑, and ←/→ buttons

1 Display the main MENU.
Press the ⊟ button to display the main MENU on your screen.

2 Select the menu you want to adjust.
Press the ↓/↑ buttons to highlight the desired menu. Press the ⊟ button to select the menu item.



3 Adjust the menu.
Press the ↓/↑ buttons to select the desired adjustment item. Press the ←/→ buttons to make the adjustment.



4 Close the menu.
Press the ⊟ button once to return to the main MENU, and twice to return to normal viewing. If no buttons are pressed, the menu closes automatically after about 30 seconds.

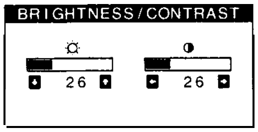
■ Resetting the adjustments

Press the →•← (reset) button. See page 14 for more information on resetting the adjustments.

Adjusting the brightness and contrast

Brightness and contrast adjustments are made using a separate BRIGHTNESS/CONTRAST menu.
These settings are stored in memory for the signals from the currently selected input connector.

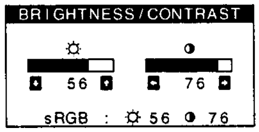
1 Press either one of the ☼ (brightness) ↓/↑ or ◑ (contrast) ←/→ buttons.
The BRIGHTNESS/CONTRAST menu appears on the screen.



2 Press the ☼ (brightness) ↓/↑ buttons to adjust the brightness (☼), and ◑ (contrast) ←/→ buttons to adjust the contrast (◑).

If you are using the sRGB mode

If you selected the sRGB mode in the COLOR menu. the following BRIGHTNESS/CONTRAST menu appears on the screen.



For more information about using the sRGB mode. see “Adjusting the color of the picture (COLOR)” on page 12.

The menu automatically disappears after about 3 seconds.

GB

Adjusting the size of the picture (SIZE/CENTER)

This setting is stored in memory for the current input signal.

1 **Press the button.**
The main MENU appears on the screen.

2 **Press the ↓/↑ buttons to highlight SIZE/CENTER and press the button again.**
The SIZE/CENTER menu appears on the screen.

3 **First press the ↓/↑ buttons to select for horizontal adjustment, or for vertical adjustment. Then press the ←/→ buttons to adjust the size.**

Adjusting the centering of the picture (SIZE/CENTER)

This setting is stored in memory for the current input signal.

1 **Press the button.**
The main MENU appears on the screen.

2 **Press the ↓/↑ buttons to highlight SIZE/CENTER and press the button again.**
The SIZE/CENTER menu appears on the screen.

3 **First press the ↓/↑ buttons to select for horizontal adjustment, or for vertical adjustment. Then press the ←/→ buttons to adjust the centering.**

Enlarging or reducing the picture (ZOOM)

This setting is stored in memory for the current input signal

1 **Press the button.**
The main MENU appears on the screen.

2 **Press the ↓/↑ buttons to highlight SIZE/CENTER and press the button again.**
The SIZE/CENTER menu appears on the screen.

3 **Press the ↓/↑ buttons to select (zoom), and press the ←/→ buttons to enlarge or reduce the picture.**

Note
Adjustment stops when either the horizontal or vertical size reaches its maximum or minimum value.

Adjusting the shape of the picture (GEOMETRY)

The GEOMETRY settings allow you to adjust the rotation and shape of the picture.
The (rotation) setting is stored in memory for all input signals. All other settings are stored in memory for the current input signal.

1 **Press the button.**
The main MENU appears on the screen.

2 **Press the ↓/↑ buttons to highlight GEOMETRY and press the button again.**
The GEOMETRY menu appears on the screen.

3 **First press the ↓/↑ buttons to select the desired adjustment item. Then press the ←/→ buttons to make the adjustment.**

Select	To
	rotate the picture
	expand or contract the picture sides
	shift the picture sides to the left or right
	adjust the picture width at the top of the screen
	shift the picture to the left or right at the top of the screen

Adjusting the convergence (CONVERGENCE)

The **CONVERGENCE** settings allow you to adjust the quality of the picture by controlling the convergence. The convergence refers to the alignment of the red. green. and blue color signals, If you see red or blue shadows around letters or lines. adjust the convergence.
These settings are stored in memory for all input signals.

1 Press the button.
The main MENU appears on the screen.

2 Press the ↓/↑ buttons to highlight CONVERGENCE and press the button again.
The **CONVERGENCE** menu appears on the screen.

3 First press the ↓/↑ buttons to select the desired adjustment item. Then press the ←/→ buttons to make the adjustment.

Select	To
	horizontally shift red or blue shadows
	vertically shift red or blue shadows
TOP V CONVER TOP	vertically shift red or blue shadows at the top of the screen
BOT **V CONVER** BOTTOM	vertically shift red or blue shadows at the bottom of the screen

Adjusting the quality of the picture (SCREEN)

The **SCREEN** settings allow you to adjust the quality of the picture by controlling the moire and landing.

- If the color is irregular at the corners of the screen. adjust the landing.
- If elliptical or wavy patterns appear on the screen, cancel the moire.

The **CANCEL MOIRE** and MOIRE ADJUST settings are stored in memory for the current input signal. All other settings are stored in memory for all input signals.

1 Press the button.
The main MENU appears on the screen.

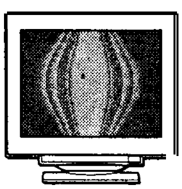
2 Press the ↓/↑ buttons to highlight SCREEN and press the button again.
The SCREEN menu appears on the screen.

3 First press the ↓/↑ buttons to select the desired adjustment item. Then press the ←/→ buttons to make the adjustment.

Select	To
LANDING	reduce any color irregularities in the screen's top left corner to a minimum.
LANDING	reduce any color irregularities in the screen's top right comer to a minimum.
LANDING	reduce any color irregularities in the screen's bottom left comer to a minimum.
LANDING	reduce any color irregularities in the screen's bottom right comer to a minimum.
CANCEL MOIRE*	turn the moire cancellation function ON or OFF. (MOIRE ADJUST) appears in the menu when you select ON.
MOIRE ADJUST	adjust the degree of moire cancellation until the moire is at a minimum.

* Moire is a type of natural interference which produces soft, wavy lines on your screen. It may appear due to interference between the pattern of the picture on the screen and the phosphor pitch pattern of the monitor.

Example of moire



Note
The picture may become fuzzy when CANCEL MOIRE is set to ON.

GB

Adjusting the color of the picture (COLOR)

The COLOR settings allow you to adjust the picture's color temperature by changing the color level of the white color field. Colors appear reddish if the temperature is low, and bluish if the temperature is high. This adjustment is useful for matching the monitor's color to a printed picture's colors.

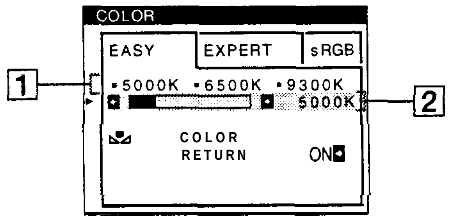
1 Press the button.
The main MENU appears on the screen.

2 Press the ↓/↑ buttons to highlight COLOR and press the button again.
The COLOR menu appears on the screen.

3 Press the ←/→ buttons to select the adjustment mode.
There are three types of adjustment modes, EASY, EXPERT and sRGB.

4 First press the ↓/↑ buttons to select the desired adjustment item. Then press the ←/→ buttons to make the adjustment.
Adjust the selected mode according to the following instructions.

EASY mode

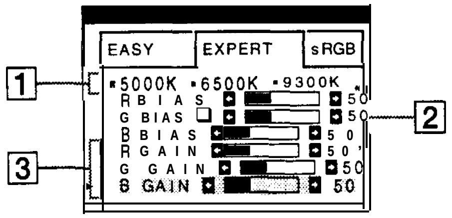


1 First press the ↓/↑ buttons to select the color temperature row [1], then press the ←/→ buttons to select a color temperature.
The preset color temperatures are 50OOK. 6500K. and 9300K. Since the default setting is 9300K, the whites will change from a bluish hue to a reddish hue as the temperature is lowered to 6500K and 50OOK.

2 If necessary, fine tune the color temperature. First press the ↓/↑ buttons to select the color temperature adjustment row [2], then press the ←/→ buttons to fine tune the color temperature.
If you fine tune the color temperature, the new color settings are stored in memory for each of the three color temperatures and item [1] of the on-screen menu changes as follows.
·[5000K]→[1]
·[6500K]→[2]
· [9300K]→[3]

EXPERT mode

You can make additional adjustments to the color in greaterdetail by selecting the EXPERT mode.



1 Press the ↓/↑ buttons to select the color temperature row [1], then press the ←/→ buttons to select a color temperature.

2 Press the ↓/↑ buttons to select the adjustment item [2], then press the ←/→ buttons to adjust the BIAS (black level).
This adjusts the dark areas of an image.

3 Press the ↓/↑ buttons to select the adjustment item [3], then press the ←/→ buttons to adjust the GAIN (white level).
This adjusts the light areas of an image.

You can adjust the R (red). G (green). B (blue) component of the input signal when making changes to items [2] and [3].

If you fine tune the color temperature, the new color settings are stored in memory for each of the three color temperatures and item [1] of the on-screen menu change as follows.
·[5000K]→[1]
·[6500K]→[2]
· [9300K]→[3]

Setting the color temperature for each of the video input connectors

You can set the fine tuning of the color temperature in EASY or EXPERT mode for each of the video input connectors (INPUT 1 and INPUT 2).

1 Select the same adjustment mode and color temperature in the COLOR menu for both INPUT 1 and INPUT 2.

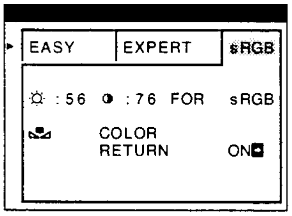
2 Fine tune the color temperature in each menu for INPUT 1 and INPUT 2.
The settings are stored in memory for each of the INPUT 1 and INPUT 2 connectors.

For information on how to select the connector, see page 7.

sRGB mode

The sRGB color setting is an industry standard color space protocol designed to correlate the displayed and printed colors of sRGB compliant computer products. To adjust the colors to the sRGB profile. simply select the sRGB mode in the COLOR menu However, in order to display the sRGB colors correctly (γ=2.2, 6500K). you must set your computer to the sRGB profile and adjust the brightness (☼) and contrast (◑) to the numbers shown in the menu. For information on how to change the brightness (☼) and contrast (◑), see page 9.

Note
Your computer and other connected products (such as a printer). must be sRGB compliant.



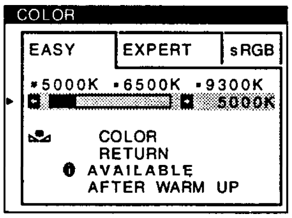
Restoring the color from the EASY or sRGB menus

The colors of most display monitors tend to gradually lose brilliance over several years of service. The COLOR RETURN feature found in the EASY and sRGB menus allows you to restore the color to the original factory quality levels. The explanation below explains how to restore the monitor's color from the EASY menu.

1 **Press the ←/→ buttons to select EASY or sRGB mode.**

2 **First press the ↓/↑ buttons to select (COLOR RETURN), then press the → button.**
The picture disappears while the color is being restored (about 2 seconds). After the color is restored, the picture reappears on the screen again.

Notes
- Before using this feature, the monitor must be in normal operation mode (green power indicator on) for at least 30 minutes, If the monitor goes into power saving mode, you must return the monitor to normal operation mode and wait for 30 minutes for the monitor to be ready. You may need to adjust your computer's power saving settings to keep the monitor in normal operation mode for the full 30 minutes. If the monitor is not ready, the following message will appear.



- The monitor may gradually lose its ability to perform this function due to the natural aging of the picture tube.

Additional settings (OPTION)

You can manually degauss (demagnetize) the monitor. change the menu position. and lock the controls.

1 **Press the ⊜ button.**
The main MENU appears on the screen.

2 **Press the ↓/↑ buttons to highlight OPTION and press the ⊜ button again.**
The OPTION menu appears on the screen.

3 **Press the ↓/↑ buttons to select the desired adjustment item.**
Adjust the selected item according to the following instructions.

Degaussing the screen

The monitor is automatically demagnetized (degaussed) when the power is turned on.
To manually degauss the monitor, first press the ↓/↑ buttons to select (DEGAUSS). Then press the → button.
The screen is degaussed for about 2 seconds. If a second degauss cycle is needed. allow a minimum interval of 20 minutes for the best result.

GB

Changing the menu' s position

Change the menu's position if it is blocking an image on the screen.
To change the menu' s on-screen position, first press the ↓/↑ buttons to select (OSD H POSITION) for horizontal adjustment, or (OSD V POSITION) for vertical adjustment. Then press the ←/→ buttons to shift the on-screen menu.

Locking the controls

To protect adjustment data by locking the controls, first press the ↓/↑ buttons to select (CONTROL LOCK). Then press the → button to select ON.
Only the (power) switch, EXIT, and (CONTROL LOCK) of the OPTION menu will operate. If any other items are selected, the mark appears on the screen.

To **cancel the control lock**
Repeat the procedure above and set (CONTROL LOCK) to OFF.

Resetting the adjustments

This monitor has the following three reset methods. Use the →•← (reset) button to reset the adjustments.

Resetting a single adjustment item

Use the ⊜, ↓/↑ buttons to select the adjustment item you want to reset, and press the →•← (reset) button.

Resetting all of the adjustment data for the current input signal

Press the →•← (reset) button when no menu is displayed on the screen.

Note that the following items are not reset by this method:

- on-screen menu language (page 7)
- adjustment mode in the COLOR menu (EASY, EXPERT, sRGB) (page 12)
- on-screen menu position (page 13)
- control lock (page 13)

Resetting all of the adjustment data for all input signals

Press and hold the →•← (reset) button for more than two seconds.

Note

The →•← (reset) button does not function when (CONTROL LOCK) is set to ON.

Technical Features

Power saving function

This monitor meets the power-saving guidelines set by VESA. ENERGY STAR. and NUTEK. If the monitor is connected to a computer or video graphics board that is DPMS (Display Power Management Signaling) compliant, the monitor will automatically reduce power consumption in three stages as shown below.

Power mode	Power consumption	⏻ (power) indicator
normal operation	≤ 145 W	green
1 standby	≤ 100 W	green and orange alternate
2 suspend (sleep)*	≤ 15 W	green and orange alternate
3 active off** (deep sleep)*	Approx. 1 W	orange
power off	0 W	off

* "Sleep" and "deep sleep" are power saving modes defined by the Environmental Protection Agency.

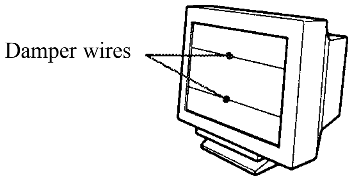
** When your computer enters a power saving mode, the input signal is cut and NO INPUT SIGNAL appears on the screen. After a few seconds, the monitor enters power saving mode.

Troubleshooting

Before contacting technical support, refer to this section.

If thin lines appear on your screen (damper wires)

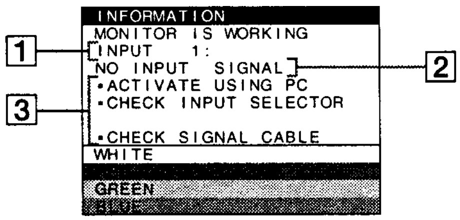
The lines you are experiencing on your screen are normal for the Trinitron monitor and are not a malfunction. These are shadows from the damper wires used to stabilize the aperture grille and are most noticeable when the screen's background is light (usually white). The aperture grille is the essential element that makes a Trinitron picture tube unique by allowing more light to reach the screen. resulting in a brighter. more detailed picture.



On-screen messages

If there is something wrong with the input signal, one of the following messages appears on the screen.

If NO INPUT SIGNAL appears on the screen



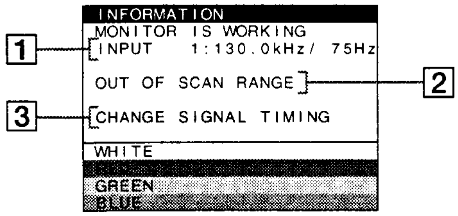
1 The selected connector
This message shows the currently selected connector (INPUT 1 or INPUT 2).

2 The input signal condition
NO INPUT SIGNAL
This indicates that no signal is input. or that no signal is input from the selected connector.

3 The remedies
One or more of the following messages may appear on the screen.

- If ACTIVATE USING PC appears on the screen. try pressing any key on the computer, and confirm that your computer's graphic board is completely seated in the correct bus slot.
- If CHECK INPUT SELECTOR appears on the screen, try changing the input signal (page 7).
- If CHECK SIGNAL CABLE appears on the screen, check that the monitor is correctly connected to the computer (page 6).

If OUT OF SCAN RANGE appears on the screen



1 The selected connector and the frequencies of the current input signal
This message shows the currently selected connector (INPUT 1 or INPUT 2). If the monitor recognizes the frequencies of the current input signal. the horizontal and vertical frequencies are also displayed.

2 The input signal condition
OUT OF SCAN RANGE
This indicates that the input signal is not supported by the monitor's specifications.

3 The remedies
CHANGE SIGNAL TIMING appears on the screen. If you are replacing an old monitor with this monitor, reconnect the old monitor. Then adjust the computer's graphic board to that the horizontal frequency is between 30 - 121 kHz, and the vertical frequency is between 48 - 160 Hz.

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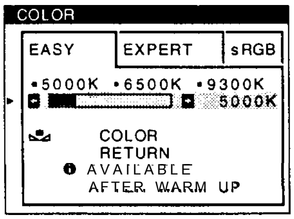
For more information, see "Trouble symptoms and remedies" on page 16.

Trouble symptoms and remedies

If the problem is caused by the connected computer or other equipment, please refer to the connected equipment's instruction manual. Use the self-diagnosis function (page 18) if the following recommendations do not resolve the problem.

Symptom	Check these items
No picture	
If the ⏻ (power) indicator is not lit	• Check that the power cord is properly connected. • Check that the ⏻ (power) switch is in the "on" position.
If the NO INPUT SIGNAL message appears on the screen, or if the ⏻ (power) indicator is either orange or alternating between green and orange	• Check that the ⇥ switch setting is correct (page 7). • Check that the HD15 video input connector's pins are not bent or pushed in. **■ Problems caused by the connected computer or other equipment** • The computer is in power saving mode. Try pressing any key on the computer keyboard. • Check that the computer's power is "on." • Check that the graphic board is completely seated in the proper bus slot.
If the NO INPUT SIGNAL and CHECK SIGNAL CABLE message appear on the screen, or if the ⏻ (power) indicator is either orange or alternating between green and orange	• Check that the video signal cable is properly connected and all plugs are firmly seated in their sockets (page 6). • Check that the HD15 video input connector's pins are not bent or pushed in. • Check that the ⇥ switch setting is correct (page 7). **■ Problems caused by the connected computer or other equipment** • Check that the graphic board is completely seated in the proper bus slot.
If the OUT OF SCAN RANGE message appears on the screen	**■ Problems caused by the connected computer or other equipment** • Check that the video frequency range is within that specified for the monitor. If you replaced an old monitor with this monitor, reconnect the old monitor and adjust the frequency range to the following. Horizontal: 30 – 121 kHz Vertical: 48 – 160 Hz
If no message is displayed and the ⏻ (power) indicator is green or flashing orange	• Use the Self-diagnosis function (page 18).
Picture flickers, bounces, oscillates, or is scrambled	• Isolate and eliminate any potential sources of electric or magnetic fields such as other monitors. laser printers, electric fans, fluorescent lighting, or televisions. • Move the monitor away from power lines or place a magnetic shield near the monitor. • Try plugging the monitor into a different AC outlet, preferably on a different circuit. • Try turning the monitor 90” to the left or right. **■ Problems caused by the connected computer or other equipment** • Check your graphics board manual for the proper monitor setting. • Confirm that the graphics mode and the frequency of the input signal are supported by this monitor. Even if the frequency is within the proper range, some video boards may have a sync pulse that is too narrow for the monitor to sync correctly. • Adjust the computer's refresh rate (vertical frequency) to obtain the best possible picture.
Picture is fuzzy	• Adjust the brightness and contrast (page 9). • Degauss the monitor* (page 13). • If CANCEL MOIRE is ON, the picture may become fuzzy. Decrease the moire cancellation effect or set CANCEL MOIRE to OFF (page 11).

Symptom	Check these items
Picture is ghosting	• Eliminate the use of video cable extensions and/or video switch boxes. • Check that all plugs are firmly seated in their sockets.
Picture is not centered or sized properly	• Press the ⊡ button (page 7). • Adjust the size (page 10) or centering (page 10). Note that some video modes do not till the screen to the edges.
Edges of the image are curved	• Adjust the geometry (page 10).
Wavy or elliptical pattern (moire) is visible	• Set CANCEL MOIRE to ON and adjust the degree of moire cancellation until the moire is at a minimum (page 1 I). ■ **Problems caused by the connected computer or other equipment** • Change your desktop pattern.
Color is not uniform	• Degauss the monitor* (page 13). If you place equipment that generates a magnetic held. such as a speaker, near the monitor, or if you change the direction the monitor faces. color may lose uniformity. • Adjust the landing (page 11).
White does not look white	• Adjust the color temperature (page 12).
Letters and lines show red or blue shadows at the edges	• Adjust the convergence (page 11).
Monitor buttons do not operate (On appears on the screen)	• If the control lock is set to ON, set it to OFF (page 13).
COLOR RETURN function does not operate  	• Before using this function. the monitor must be in normal operation mode (green power indicator on) for at least 30 minutes. For more information on using the COLOR RETURN function. see page 13. • Adjust the computer's power saving settings to keep the monitor in normal operation mode for more than 30 minutes. • The monitor may gradually lose its ability to perform this function due to the natural aging of the picture tube.
A hum is heard right after the power is turned on	• This is the sound of the auto-degauss cycle. When the power is turned on. the monitor is automatically degaussed for two seconds.

* If a second degauss cycle is needed, allow a minimum interval of 20 minutes for the best result. A humming noise may be heard, but this is not a matfunction.

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Displaying this monitor' s name, serial number, and date of manufacture.

While the monitor is receiving a video signal, press and hold the ⊟ button for more than five seconds to display this monitor's information box.

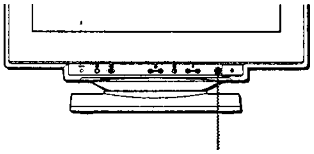
Example

If the problem persists, call your service representative and give the following information.

- Model name: GDM-5402T9
- Serial number
- Name and specifications of your computer and graphics board.

Self-diagnosis function

This monitor is equipped with a self-diagnosis function. If there is a problem with your monitor or computer(s), the screen will go blank and the ⏻ (power) indicator will either light up green or flash orange. If the ⏻ (power) indicator is lit in orange. the computer is in power saving mode. Try pressing any key on the keyboard.



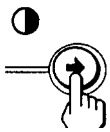
⏻ (power) indicator

If the ⏻ (power) indicator is green

1 **Remove any plugs from the video input 1 and 2 connectors, or turn off the connected computer(s).**

2 **Press the ⏻ (power) button twice to turn the monitor off and then on.**

3 **Press the ➡ button for 2 seconds before the monitor enters power saving mode.**



If all four color bars appear (white, red. green, blue), the monitor is working properly. Reconnect the video input cables and check the condition of your computer(s).

If the color bars do not appear, there is a potential monitor failure. Inform your service representative of the monitor's condition.

If the ⏻ (power) indicator is flashing orange

Press the ⏻ (power) button twice to turn the monitor off and then on.
If the ⏻ (power) indicator lights up green. the monitor is working properly.

If the ⏻ (power) indicator is still flashing. there is a potential monitor failure. Count the number of seconds between orange flashes of the ⏻ (power) indicator and inform your service representative of the monitor's condition. Be sure to note the model name and serial number of your monitor. Also note the make and model of your computer and video board.

Specifications

CRT	0.24 mm aperture grille pitch 21 inches measured diagonally 90-degree deflection FD Trinitron
Viewable image size	Approx. 403.8 x 302.2 mm (w/h) (16 x 12 inches) 19.8" viewing image
Recommended resolution	Horizontal: 1600 dots Vertical: 1280 lines
Standard image area	Approx. 388 x 291 mm (w/h) (15 3/8 x 11 1/2 inches) or Approx. 364 x 29 1 mm (w/h) (14 3/8 x 11 1/2 inches)
Deflection frequency*	Horizontal: 30 to 121 kHz Vertical: 48 to 160 Hz
AC input voltage/current	100 to 240 V, 50 – 60 Hz, 2.0 – 1.0 A
Power consumption	Approx. 145 W
Dimensions	Approx. 498 x 491 x 478 mm (w/h/d) (19 5/8 × 19 3/8 × 18 7/8 inches)
Mass	Approx. 32 kg (70 lb 9 oz)
Plug and Play	DDC1/2B/2Bi, GTF**

* Recommended horizontal and vertical timing condition
- Horizontal sync width duty should be more than 4.8% of total horizontal time or 0.8 µs, whichever is larger.
- Horizontal blanking width should be more than 2.3 µsec.
- Vertical blanking width should be more than 450 µsec.

** If the input signal is Generalized Timing Formula (GTF) compliant, the GTF feature of the monitor will automatically provide an optimal image for the screen.

Design and specifications are subject to change without notice.